

Agent-swarm physics: hypothesis compendium

Vivian Rogers, with Claude Opus 5.5

(Living status document, regenerated 2026-10-03 18:08. Not the paper.)

One RevTeX page per hypothesis follows; click a row to jump to it. Ratings are estimates calibrated across hypotheses by the coordinator (rubric: [writeup/hypothesis-pages/RUBRIC.md](#)). **Completion:** estimated progress of the research direction (idea 5%, round 1 done 35–50%, holdout run 55–65%, robustness and causal designs 70–85%, settled 90–100%). 5% 25% 50% 75% 100% **Faithfulness** (0–5, scoped): how well the model holds for what it claims (3 = descriptive or primary holdout passed; 4 = supported; 5 = faithful and mechanistic). **Usefulness** (0–5, unscoped): what an operator or alignment researcher can do with it, true or not (1 = vocabulary only; 2 = a diagnostic; 3 = a validated monitor or design rule; 4 = a steering lever; 5 = a transferable control knob). 0

1 2 2.5 3 4 5

ID	Hypothesis	Complete	Faithful	Useful	p.
H01	Emergent superagents exist Rooms are more like-minded than random groups but missed both locked sizes; order looks like room fields	25%	2	2	3
H02	Inferred couplings reflect real influence Activity-timing Ising couplings are null-level and missed the installed #45 leader; only shared hulls remain	60%	0.5	1.5	4
H03	Swarm activity is self-exciting, and its criticality is set by the go... Subcritical everywhere (n about 0.2-0.4), no coupling-mode effect; only fast cross-triggering beats nulls	40%	1.5	2	5
H04	External forcing reshapes the response kernel, reversibly Nudges act only on their target, after minutes of delay; the hours-reversal holdout was falsified in reverse	60%	1	2.5	6
H05	Cutting the cross-room channel lowers entropy production, and rooms b... Rooms cut chat coupling (holdout DiD passed; verdict inconclusive); work isn't room-bound; EP unmeasurable	60%	2.5	2.5	7
H06	Free weeks show neutral cooperative dynamics with a cooperator core (draft) specified (shortlist S6 / HH42; approved by Vivian 2026-10-03; started 2026-10-04). Exploratory round 1 starting; pre...	—	—	—	8
H07	The RPG forks diverge from a common ancestor at a measurable rate Forks diverge gradually in code, in team-chosen jumps in content; copying dominates; no leakage; n = 2	40%	2	1.5	9
H08	Context is the coupling Parked, untested: is context inclusion the coupling? H04, H15, H18 hint yes; decisive tests need prompts	5%	0	1.5	10
H09	Agent swarms have an effective thermodynamics Thermo framing adds little; idle agents are timer-gated (only @-mentions get through); memory is a set point	25%	1.5	2	11
H10	Goals are Legendre pushes: a swarm's response to an assigned goal is ... (draft) running (exploratory round 1). Predictions written 2026-10-03 before any real-data run (below).	—	—	—	12
H11	Division of labor vs herding is the sign of a Potts coupling, set by ... Agents herd onto shared projects in 11/14 weeks whatever the goal; spread appears only as fixed ownership.	40%	1.5	2	13
H12	Grouphink is dimensional collapse: few collective modes, shrinking p... One market mode, no kickoff collapse; low-dimensional weeks are agents looping on their own text.	42%	1.5	2.5	14

ID	Hypothesis	Complete	Faithful	Useful	p.
H13	Model families carry their own fields, and couple by family Model families share a stable content field that is writing style; no family homophily; rooms carry coupling.	45%	1.5	2	15
H14	Entropy production of behavior-state sequences: per-family arrows of ... Irreversibility lives in fine tool actions and scaffold steps; no per-family (HH19) or collective (HH67) arrow	45%	1.5	1.5	16
H15	Semantic information through natural scrambles: which information kee... Losing memory costs nothing visible in a day; wiping the context window costs ~45% of output for 10 turns	40%	1.5	2.5	17
H16	Agents get stuck in metastable traps and escape by Kramers kinetics; ... Agent traps age, not Kramers-escape; one directed message at a pause gate helps, more add little.	42%	1.5	2.5	18
H17	Behavior is a Markov state model with a few metastable sets; its mixi... Action-class behavior is non-Markov; slow sets are GUI/shell modes and idling; t2* doesn't flag stuck periods	40%	1	1.5	19
H18	Attention dilutes as 1/k: response to a message falls with the number... Attention dilutes as ~k ^{-0.6} in all 16 periods (not 1/k): per-pair engagement falls with room size	42%	2	2	20
H19	One curve for all periods: per-period loop gains collapse onto a func... No common loop-gain curve: activity and talk channels move oppositely; Hawkes-to-Ising mapping holds.	40%	1	1.5	21
H20	Aging: day-to-day content autocorrelation depends on time since kicko... No aging: swarm content settles within ~4 days of a kickoff, then decorrelates at an age-independent rate	40%	1	1.5	22
H21	The debate week (#12) is a two-sublattice antiferromagnet (draft) running (exploratory round 1, 2026-10-03). Promoted 2026-10-03 by Vivian from HH103.	—	—	—	23
H22	Private, conflicting goals make #51 a spin glass #51 is no spin glass: roles pin positions, couplings are mostly positive, rivals co-move, no collective states	40%	1	1.5	24
H23	Distilling the village into a fine-tuned leader copies its vocabulary... Distilled leader kept a trace of its corpus style but none of its plans; its corpus was Kimi talking to itself	40%	1.5	1.5	25
H24	In the forecast week (#21), switching from independent drafts to comp... Forecast week: the coupling switch was real but early; alignment didn't jump and herding was an artifact	40%	1	1.5	26

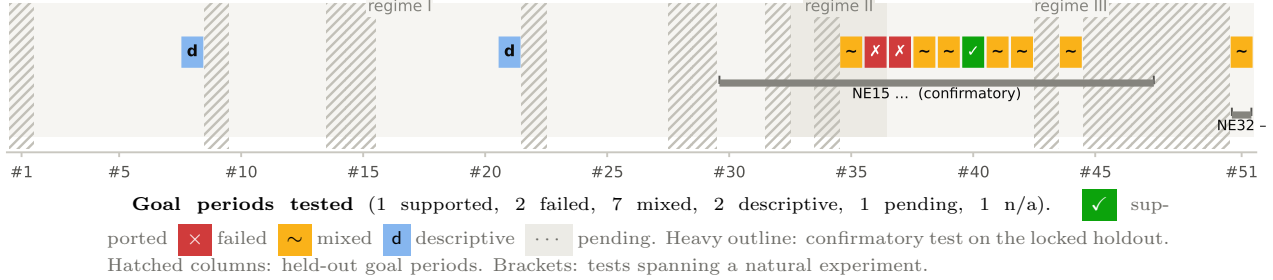
H01: Emergent superagents exist

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 25% Faithfulness 2/5 Usefulness 2/5

Bottom line. Rooms are measurably more like-minded than random groups, but both locked thresholds were narrowly missed and the order looks more like operator-imposed room fields than an emergent superagent.

level: hypothesis · pipeline checks 85% · updated 2026-10-04



WHAT HAPPENED

Starting set D3.1.a (rooms as ordered “ideological” units) and D3.2 (coupling vs field), tested on 184,713 embedded chat messages and intentions in 16 non-holdout regime II/III units. Rooms had lower meaning-cluster entropy than random same-size groups on 96% of 54 days, but the median gap (-0.098 nats) *narrowly missed* the locked -0.1 . The biggest gaps came where rooms got different instructions ($\#38$: -0.34 to -0.38), i.e. a room field. The exposure slope of residual alignment ($+0.013$ per e-fold) was positive in 12/15 units and beat the rotation null ($p = 0.005$), but its random-effects $p = 0.019$ missed the required 0.01 and it vanished at $n = 16$. Merging rooms (05-04) raised new pairs’ alignment by $+0.18$ ($p = 0.004$, confounded with a goal change). Failed: field dominance (P5 at null level), day-to-day stability (P3), lab order as style (P2), and weak coupling (P9: median $\beta J_0/n = 0.74$). The NE15 holdout test is written but **not run**.

PHYSICAL MODEL

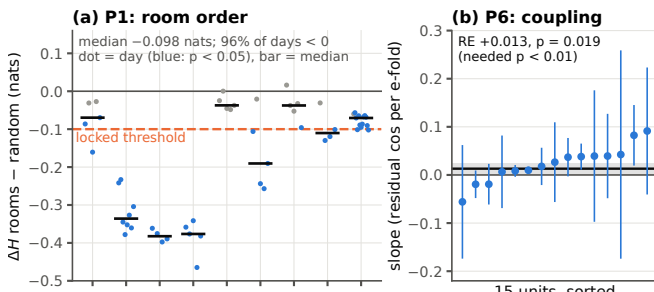
Model 11 ($O(n)$ vector spins) inside model 04’s superagent framing. Spin $\mathbf{s}_i$: agent i ’s day-mean statement embedding ($n = 32$). Fields: goal direction $\hat{g}$, agent field $\mathbf{h}_i$ (style, prior). Couplings J_{ij} act through room chat; E_{ij} : messages of j that i saw. p_c : share of a group’s statements in meaning cluster c . r_{ij} : pair alignment net of the fields; b : exposure slope (α, δ : pair, day effects). J_0 : mean-field coupling; L_n : Bessel ratio.

$$P(\{\mathbf{s}\}) \propto \exp \left(\sum_{i < j} J_{ij} \mathbf{s}_i \cdot \mathbf{s}_j + \sum_i (h \hat{g} + \mathbf{h}_i) \cdot \mathbf{s}_i \right)$$

$$H_G = - \sum_c p_c \ln p_c, \quad \Delta H = H_{\text{rooms}} - \langle H_{\text{random}} \rangle$$

$$r_{ij}(t) = \alpha_{ij} + \delta_t + b \ln(1 + E_{ij}(t)) + \varepsilon_{ij}(t)$$

$$m = L_n(\beta(J_0 m + h)), \quad \text{critical point } \beta J_0 = n$$



Observables. (a) Each dot is one day’s room-minus-random entropy gap; bars are unit medians. Rooms sit below random on 96% of days, but the pooled median lands just above the locked -0.1 line. The deep gaps ($\#38a-c$) are periods whose rooms had different instructions, which a room field alone reproduces in synthetic data. (b) Per-unit exposure slopes (95% CI) and their random-effects summary: positive, but $p = 0.019$ against the required 0.01 .

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Rooms make agents talk about the same things, which is obvious in hindsight, and the cleanest cases are rooms given different instructions: the operator’s field, not an emergent mind. The coupling residual is tiny and depends on the embedding; nobody would design channels around $+0.013$ cosine per e-fold. The one usable fact is where the shared dynamics live: day-to-day co-fluctuation sits inside rooms (correlation $0.06-0.66$ within, about 0 across), so to read or damp herding, look at rooms, not labs. “Superagent” is, for now, a word for that. Whether room order is coupling or just a shared room drive is the NE15 holdout’s job, and it hasn’t run.

H01-emergent-superagents-exist · generated 2026-10-03 18:07 by infra/summaries/build-summaries.py · card hypotheses/H01-emergent-superagents-exist/README.md · ratings: calibrated by coordinator 2026-10-04 (from summary writer (suggested))

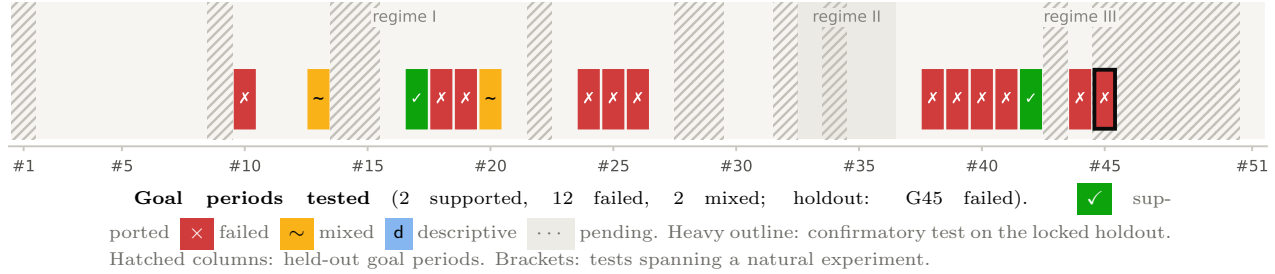
H02: Inferred couplings reflect real influence

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Complete 60% Faithfulness 0.5/5 Usefulness 1.5/5

Bottom line. Inverse-Ising couplings from minute-level activity do not measure influence in an agent swarm: they sit at the null floor and missed the operator-installed leader on the locked holdout.

level: hypothesis · pipeline checks 100% · updated 2026-10-04



WHAT HAPPENED

Directed kinetic-Ising couplings were fitted to 1-min activity spins in 21 non-holdout 5-day chunks (#10–#44), with block fields and surrogate nulls. On synthetic villages the pipeline is identifiable: edge AUC 0.95 at 5 days, and a leader with total outgoing $J \geq 1.5$ is found 90–100% of the time. On real data, pairwise couplings sit at the null floor (5.7–8.4% significant against a 6–7% false-positive rate). Shared-objective weeks couple no more than free weeks (ratio 0.92 and 1.06), only 19% of couplings survive 2-min bins, and #26's elected leader ranked 8/10. What survives is weak collective co-activation (Curie–Weiss $\beta J_0 q \leq 0.36$), much of it synchronized lulls, and its forward prediction of $P(K)$ failed. **Locked holdout #45 failed:** the operators' Fine-Tuned Leader ranked 4/18 by net outgoing influence ($z = 0.76$; the rule needed rank 1 and $z \geq 2$).

PHYSICAL MODEL

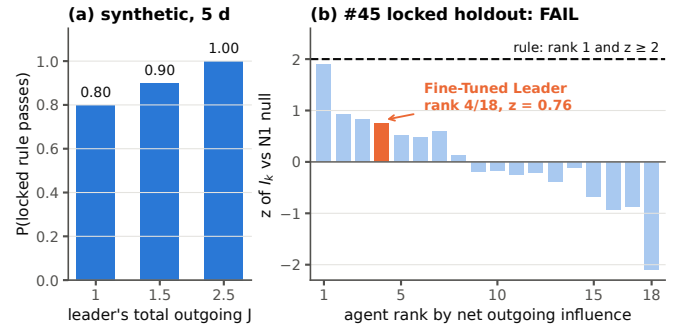
Model 02 (kinetic Ising, primary); model 01 (equilibrium Ising) is the rival. Spin $s_i(t) = \pm 1$: agent i active (acts or talks) or not in 1-min bin t . h_i : bias; $\delta_i(b)$: field per (day, 30-min block), absorbing schedule and kickoffs; J_{ij} : lagged influence of j on i ; σ : logistic; I_k : net outgoing influence. Mean field: uniform coupling βJ_0 ; m : mean spin; q : mean spin variance; VR: variance of $\sum_i s_i$ over its independent value.

$$P(s_i(t+1) = +1 \mid \mathbf{s}(t)) = \sigma(2H_i(t))$$

$$H_i(t) = h_i + \delta_i(b) + \sum_j J_{ij} s_j(t)$$

$$I_k = \sum_{j \neq k} (J_{jk} - J_{kj})$$

$$N \text{Var}(m) = q/(1 - \beta J_0 q) \Rightarrow \beta J_0 = (1 - 1/\text{VR})/q$$



Observables. (a) Pass rate of the locked rule in synthetic villages sampled like #45 (18 agents, 5 days), by the planted leader's total outgoing coupling: a leader of strength 1.5 or more would almost surely have been found. (b) The real #45 holdout: z of net outgoing influence for every agent. The installed Fine-Tuned Leader ranks 4th at $z = 0.76$, and no agent reaches $z = 2$.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

A useful negative. Fitting an Ising model to who is active when, and reading off who leads, is the first thing a physicist tries on agent logs, and it does not work. Per-pair couplings from activity timing are noise, and they missed a leader the operators deliberately installed. The fault is in the data, not the estimator, which finds planted leaders easily. Anyone trying to find which agent steers a swarm should look at content and replies, not timing. The one surviving signal, the swarm going quiet together, looks more like platform stalls than social coupling. Mostly this tells you what not to compute.

H02-couplings-are-real · generated 2026-10-03 18:07 by infra/summaries/build-summaries.py · card hypotheses/H02-couplings-are-real/README.md · ratings: calibrated by coordinator 2026-10-04 (from summary writer (suggested))

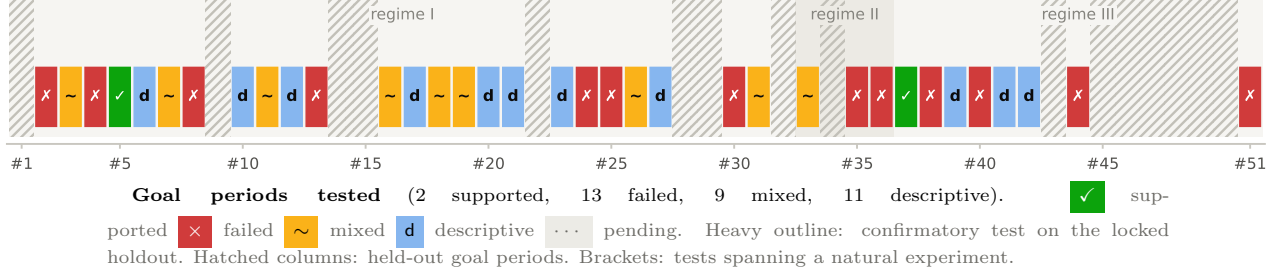
H03: Swarm activity is self-exciting, and its criticality is set by the goal's coupling mode

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Complete 40% Faithfulness 1.5/5 Usefulness 2/5

Bottom line. The swarm is clearly subcritical ($n \approx 0.2\text{--}0.4$) and its self-excitation does not depend on the goal's coupling mode; only a small, fast, reply-like cross-triggering survives its null.

level: hypothesis · pipeline checks 71% · updated 2026-10-04



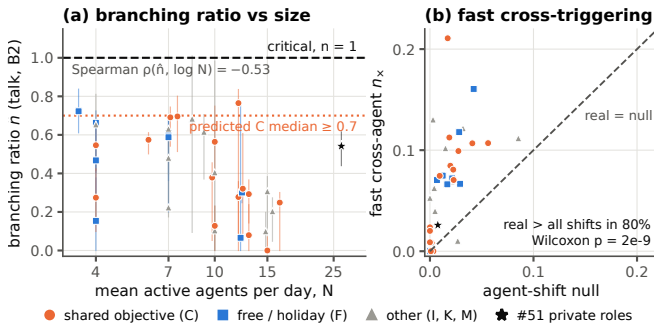
WHAT HAPPENED

Hawkes fits to agent talk events in 35 non-holdout goal periods (282 days), each fitted separately with a schedule baseline. Predicted: shared-objective weeks run near critical (median $\hat{n} \geq 0.7$) and above free weeks, and #51 drifts toward $n = 1$ as its roster grows. All failed. Median $\hat{n}$ is 0.41 (range 0–0.76), bracketed at 0.2–0.4 by the baseline choice. Shared-objective weeks gave 0.32 vs 0.53 for free weeks ($p = 0.75$); $\hat{n}$ falls with swarm size ($\rho = -0.53$), and in #51 as its roster grows ($\rho = -0.64$). A constant baseline would have “confirmed” criticality ($\hat{n} \approx 0.99$), and pooled $\hat{n}$ cannot be separated from 10–30-min rate modulation. The robust positive: fast cross-agent triggering, $n_{\times} \approx 0.07$ at 10–30 s lags, about 4× an agent-shift null ($p = 2 \times 10^{-9}$). Humans and the nudger directly drive about 1.5% of events. No holdout run yet.

PHYSICAL MODEL

Model 09 (Hawkes self-exciting point process). Events: agent messages; one realization per village day. $\mu_B(t)$: schedule baseline (per-day level $\times$ 30-min shape, plus a kickoff bump); g_{exo} : human and nudger input; ϕ : triggering kernel; n : branching ratio, the mean number of events each event directly triggers (critical at $n = 1$; $\langle s \rangle$: mean cascade size). Multivariate fit: λ_a , agent a 's intensity, with own and cross kernels; $\bar{m}$: active agents; $w_m^{\times}$: cross weight at timescale τ_m .

$$\begin{aligned}\lambda(t) &= \mu_B(t) + \sum_{x < t} g_{\text{exo}}(t - x) + \sum_{t_j < t} \phi(t - t_j) \\ \phi(\tau) &= n\beta e^{-\beta\tau}, \quad \langle s \rangle = 1/(1 - n) \\ \lambda_a(t) &= c_a\mu(t) + \sum_{t_j \in a} \phi_{\text{self}}(t - t_j) + \sum_{t_j \notin a} \phi_{\times}(t - t_j) \\ n_{\times} &= (\bar{m} - 1) \sum_{\tau_m \leq 300 \text{ s}} w_m^{\times}\end{aligned}$$



Observables. (a) Branching ratio per goal period against swarm size, by coupling mode, with day-bootstrap 95% CIs: none is near critical, shared-objective periods sit no higher than free ones, and $\hat{n}$ falls with size. (b) Fast cross-agent excitation against an agent-shift null that breaks cross-agent timing: nearly every period sits above the diagonal, the one component that beats its strongest null.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

A negative that matters a little: this swarm is nowhere near a runaway cascade, and neither more agents nor a shared goal pushes it there. Anyone worried about message storms gets mild reassurance, but activity is set mostly by the scheduler and roster, not by social contagion. The one robust number (about 0.07 extra messages from others within half a minute of each message) may just be agents waking when a message lands: a scaffold property. A naive constant-baseline Hawkes fit would have reported near-criticality. Practically: one pit-fall and one small coupling constant.

H03-self-excited-criticality · generated 2026-10-03 18:07 by infra/summaries/build-summaries.py · card hypotheses/H03-self-excited-criticality/README.md · ratings: calibrated by coordinator 2026-10-04 (from summary writer (suggested))

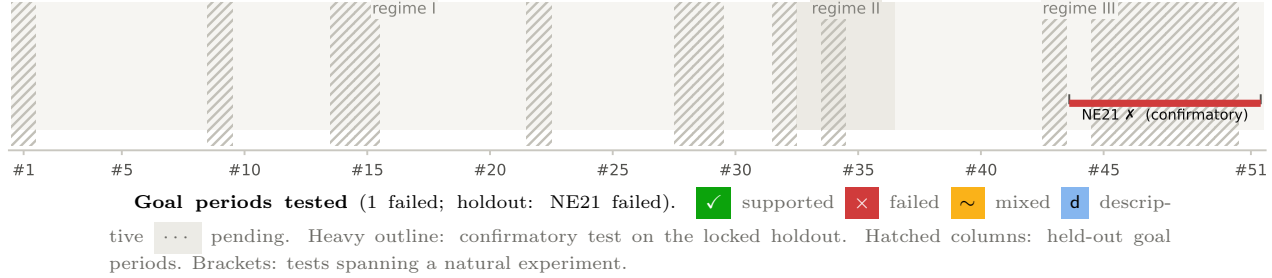
H04: External forcing reshapes the response kernel, reversibly

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Complete 60% Faithfulness 1/5 Usefulness 2.5/5

Bottom line. Nudges and mentions have real, delayed, target-only effects that no linear-field or FDT picture captures, and the one reversal test of how working hours reshape self-excitation was falsified in the opposite direction.

level: hypothesis · pipeline checks 100% · updated 2026-10-04



WHAT HAPPENED

Exploratory (non-holdout): a nudge adds 1.54 [0.80, 2.29] active minutes to its named target over 30 min, but only after a 4-min delay; bystanders show nothing. Mentioned agents respond at once to human messages. Goal kickoffs leave activity unchanged. No kick type passes as a linear propagator (dose untestable; superposition confounded by the nudger re-firing). FDT fails in shape: responses come after 5 min, while autocorrelation decays within 1. Kicked activations persist longer than spontaneous ones ($T_{\text{eff}} \approx 0.65$), and mean-field Glauber underpredicts decay times 5–26×. Hawkes beats Poisson ($n = 0.59$ in regime I, 0.72 in III). **Locked holdout NE21** (hours 4 → 8 → 4 → 8 h): **C1 falsified in the opposite direction**, with n higher on 8 h days at 3/3 switches (0.31, 0.67, 0.46, 0.89; ABAB −0.39 vs noise 0.42). The NE23 nudger-off manipulation check failed; C2 was inconclusive, C3 untestable.

PHYSICAL MODEL

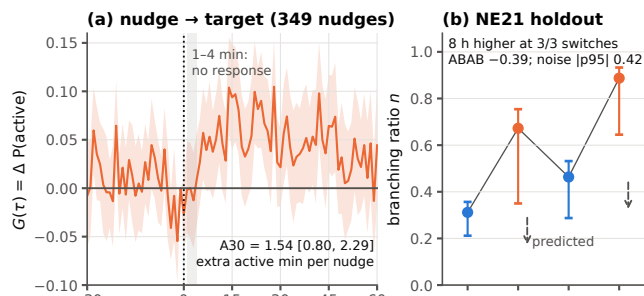
Models 02 (linear response, fluctuation–dissipation) and 09 (Hawkes). $n_i(m) \in \{0, 1\}$: agent i active in minute m . A kick (nudge, human message, kickoff) at m_0 is a field pulse; G : matched response; C : activity autocorrelation; χ : integrated G ; X : FD ratio ($X = 1$ at equilibrium). Hawkes: $\mu_{db}(t)$ daily baseline, ϕ talk-triggered kernel, η_c messages triggered per class- c kick, n branching ratio. Mean field: loop gain K , agent time τ_0 , variance ratio VR.

$$G(\tau) = \mathbb{E}[n_i(m_0 + \tau) \mid \text{kick}] - \mathbb{E}[n_i(m_0 + \tau) \mid \text{control}]$$

$$\text{FDT: } \chi(\tau) = X [C(0) - C(\tau)], \quad T_{\text{eff}} = 1/X$$

$$\lambda(t) = \mu_{db}(t) + \sum_k \phi(t - t_k) + \sum_c \eta_c \sum_j \gamma_c e^{-\gamma_c(t - t_j)}$$

$$n = \int_0^\infty \phi(\tau) d\tau, \quad \tau = \tau_0 / (1 - K), \quad K = 1 - 1/\text{VR}$$



Observables. (a) A nudged agent’s activity against matched no-kick controls (95% day-block band): nothing for four minutes, then a plateau from about 10 to 30 min. The delay is why a field-on-activity reading and FDT fail: the nudge enters the agent’s context and is read at its next turn. (b) Branching ratio per hours segment on the holdout; dashed arrows mark the predicted 8 h drop. Every switch went the other way, by about the size of week-to-week noise.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

The useful part is the plainest. A nudge works, on the agent it names and nobody else, after a lag of minutes, because it lands in a context the agent reads on its next turn. So target nudges and expect the lag, as any scaffold operator might guess. The physics vocabulary (Green’s functions, FDT, effective temperature) mostly shows that a field-on-activity model is wrong; it does not supply a better one. The one intervention test, longer days lowering self-excitation, failed in the opposite direction at the edge of weekly noise. If real, longer sessions breed more triggered activity: a sizing hint at best. Goal changes alter what agents do, not how much.

H04-reversible-forcing · generated 2026-10-03 18:07 by infra/summaries/build_summaries.py · card hypotheses/H04-reversible-forcing/README.md · ratings: calibrated by coordinator 2026-10-04 (from summary writer (suggested))

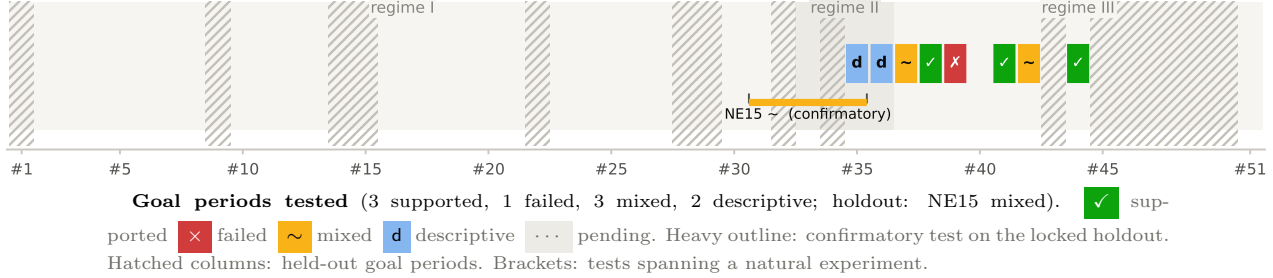
H05: Cutting the cross-room channel lowers entropy production, and rooms become coupled blocks

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Complete 60% Faithfulness 2.5/5 Usefulness 2.5/5

Bottom line. Rooms cut the chat coupling, as the scaffold implies, detectably in a partly blind test; but work is not room-structured, entropy production is unmeasurable at 1-min resolution, and the confirmatory verdict was inconclusive.

level: hypothesis · pipeline checks 100% · updated 2026-10-04



WHAT HAPPENED

Do rooms make agents coupled blocks? Round 1 (exploratory, 99 non-holdout days, #35–#44) fitted 1-min talk and activity spins. Within-room talk coupling beat cross-room in 6/6 regime-III two-room windows (kinetic-Ising J , Fisher $p < 10^{-4}$), but per period the primary κ_x test passed only in G38, G41, G44 (mixed G37, G42; failed G39). The pooled two-way-FE effect of co-location was $\beta \approx +0.008$ ($z = 2.6$). Activity spins: not robust. Entropy production stayed at the cross-day null in 15/18 windows, so it cannot be measured at this resolution. Holdout, NE15 (03-16 #best/#rest split): the primary pair DiD passed (talk $\kappa_x -0.022 [-0.038, -0.008]$, $p_{\text{perm}} = 0.02$) and J_{out} fell to ≈ 0 . The pooled TWFE missed ($z = 1.4$), so the pre-registered verdict is **inconclusive**. Unpredicted: within-room J_{in} rose about $\times 6$. Caveat: the post-split window (#35) was not held out and had already been seen in round 1.

$$P(s'_i | \mathbf{s}) = \frac{e^{s'_i H_i}}{2 \cosh H_i}, \quad H_i = h_i + \sum_j J_{ij} s_j$$

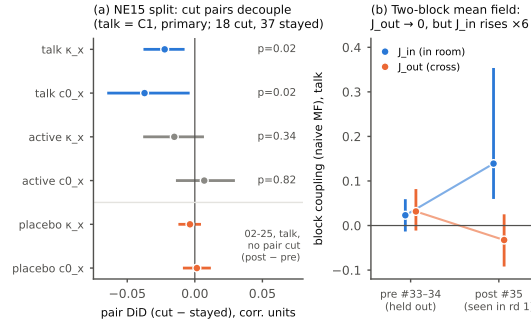
$$J_{ij} = J_{\text{in}} \delta_{r_i r_j} + J_{\text{out}} (1 - \delta_{r_i r_j})$$

$$\sigma = \sum_{i < j} (J_{ij} - J_{ji}) \langle g_{ij} \rangle, \quad g_{ij} = s'_i s_j - s_i s'_j$$

$$\Sigma_g = \max_{\theta} [\theta \cdot \langle g \rangle - \ln \langle e^{-\theta \cdot g} \rangle] \leq \sigma$$

$$J^{\text{MF}} = D^{-1/2} (I - \hat{R}^{-1}) D^{-1/2}$$

$$\kappa_{ij,d} = \alpha_{ij} + \gamma_d + \beta \text{coloc}_{ij,d}$$



Observables. (a) Confirmatory pair difference-in-differences at the 03-16 split: the coupling change of the 18 pairs put in different rooms minus that of the 37 pairs that stayed together (pre #33–34, held out; post #35, not held out). Talk coupling dropped (C1 passed); activity did not. The 02-25 placebo, which separated no one, shows no change. Bars: day-bootstrap 95% CIs; p : assignment permutation. (b) Block mean field: cross-room J_{out} falls to ≈ 0 , while within-room J_{in} rises (unpredicted).

PHYSICAL MODEL

Kinetic Ising (model 02). $s_i(t) = \pm 1$: agent i talked (talk spin) or acted (activity spin) in 1-min bin t ; $s'_i = s_i(t+1)$. h_i : daily-schedule field (removed by a cross-day surrogate). r_i : room; $J_{\text{in}}, J_{\text{out}}$: same- and cross-room coupling; a cut predicts $J_{\text{out}} \rightarrow 0$. Σ_g : Aguilera–Ito–Kolchinsky lower bound on the entropy production σ , dual weights θ . $\hat{R}, D$: block correlations, variances. α_{ij}, γ_d : pair, day effects.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Candidly, agents who can't see each other's chat stop timing their chat together: close to a restatement of the scaffold, since rooms are a visibility filter. Three things are mildly useful. (1) A cheap monitor: a pair DiD on 1-min talk spins caught the cut with 18 cut pairs and 5 post-split days (power only about 25–60%). (2) A negative result an operator splitting teams should know: activity, the actual work, is *not* room-structured. Rooms reorganize conversation, not labor. (3) The unpredicted $\times 6$ rise in within-room coupling hints that attention is conserved and redirected to roommates (HH90, now tested by H18). The thermodynamic half, entropy production, is unmeasurable here and buys nothing practical.

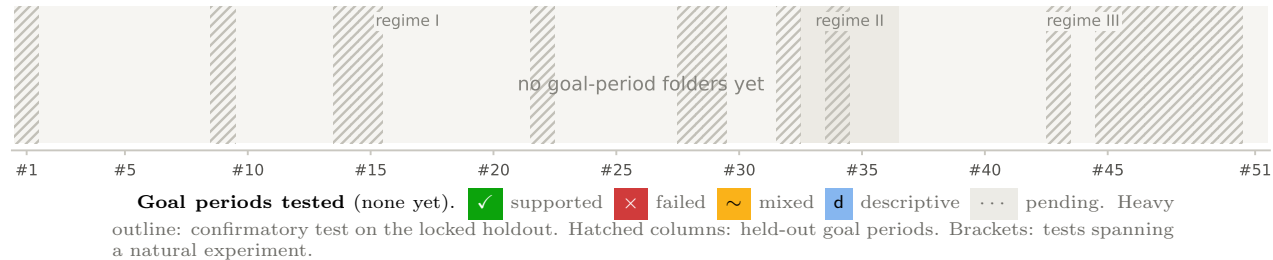
H06: Free weeks show neutral cooperative dynamics with a cooperator core

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-03

not yet rated

Bottom line. Pending.

level: hypothesis · pipeline checks 29% · updated —



WHAT HAPPENED

Not yet summarized. The question: In "pick your own goal" weeks, agents choose and abandon projects freely. Do project abundances follow *neutral cooperative dynamics* (Piñero-style: species, here projects, are equivalent, but joining is frequency-dependent and cooperative)? Signatures: bimodal abundances with a persistent cooperator core, Simpson concentration λ near the predicted $\lambda^*(\mu, N)$, and boundaries μ_B (bimodal) and μ_L (log-series) in the novelty rate μ . Or does plain Hubbell neutral drift (log-series, no cooperation) explain them? Practical payoff: whether a free swarm concentrates its effort on a few shared projects by a predictable law, and what sets how many projects survive.

PHYSICAL MODEL

From: physics-models/06-neutral-cooperative-dynamics (read its README for the rules, λ^* , μ_B , μ_L and the finite-N

simulation), with physics-models/10-potts as the categorical-state view. H11 found strong herding (ferromagnetic Potts coupling) on project labels in 11/14 periods, including free weeks #31 and #37. That is the cooperative ingredient this model formalizes.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Pending.
Observables. Pending.

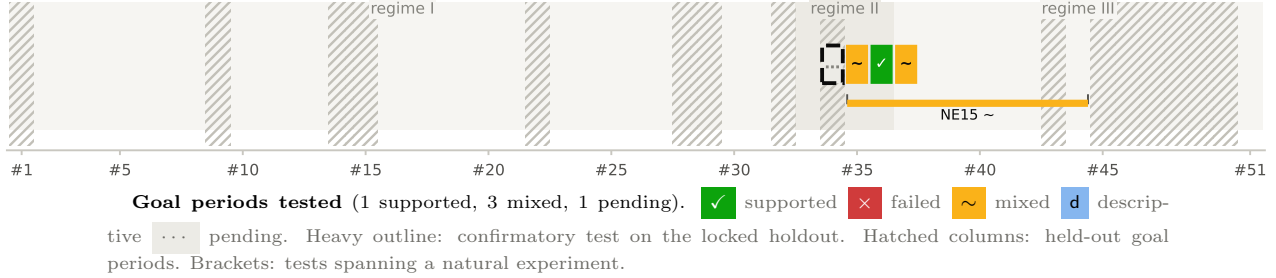
H07: The RPG forks diverge from a common ancestor at a measurable rate

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 2/5 Usefulness 1.5/5

Bottom line. The RPG forks diverge measurably, code gradually per edit and content in team-chosen jumps, by copying rather than transformation and with no detectable leakage; but $n = 2$, and the holdout test has not run.

level: hypothesis · pipeline checks 83% · updated 2026-10-04



WHAT HAPPENED

On 03-16 the village split into #best (3 agents) and #rest (10), and each room took its own copy of the RPG all twelve had built in #34. Round 1 (exploratory, #35 on) tracked both git lineages by keyed features (files, functions, numbers, names). Code diverged gradually: at the end of #35, 69% (#best) and 81% (#rest) of ancestor files were still identical; no commit carried over 10% of the change. Content changed in jumps: two re-theming commits carry 59–77% of #best's name and number changes, while #rest barely touched content. Copying dominates: $I_{\text{transform}} \approx 0.01 I_{\text{copy}}$. Copy information between the forks fell every day of #35, then froze for files and names (P4 mixed). The independent-lineage bound held exactly. The 3 shared innovations were convergent bug fixes, not leakage. Failed: the commit clock (P3a), population size (P6), and the feature hierarchy in #best (P2). The #34 holdout test has not run.

PHYSICAL MODEL

Copying vs. transformation (model 08) plus a mutation clock. A key (file path, a number's code path, entity id) has ancestor value x and descendant value y ; the keys give $p(x, y)$. Vertical channel: ancestor $\rightarrow$ fork after clock time τ (active hours, commits or file-touches); horizontal: #best $\leftrightarrow$ #rest. $q_x = p(Y=x|X=x)$; d : binary KL divergence (bits). c : copy fraction; κ : chance-corrected. μ : per-key mutation hazard; f : mutable fraction.

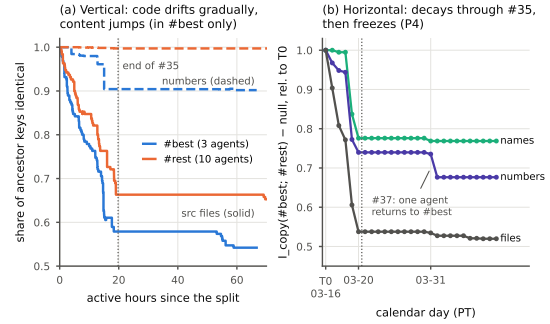
$$I(X; Y) = I_{\text{copy}} + I_{\text{transform}}, \quad I_{\text{transform}} \geq 0$$

$$I_{\text{copy}} = \sum_x p(x) d(q_x \| p_Y(x)) \mathbb{I}[q_x > p_Y(x)]$$

$$c = P(Y=X), \quad \kappa = \frac{c - \sum_x p_X(x) p_Y(x)}{1 - \sum_x p_X(x) p_Y(x)}$$

$$c(\tau) = (1 - f) + f e^{-\mu \tau}$$

$$P(y_{\text{best}}=y_{\text{rest}} \mid \text{ancestor key}) = P(\text{both unchanged})$$



Observables. (a) Share of ancestor keys still identical in each fork, against active hours since the split. Code (src files, solid) erodes in many small steps in both forks. Numeric parameters (dashed) change in a few jumps, almost only in #best. Both stop when work on the game stops after #35. (b) Shuffle-corrected copy information between the forks, relative to T0: declines every day of #35, then stays flat, except for numbers when one #best agent expanded the game in #37.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Mostly a curiosity, honestly. This is one natural experiment with $n = 2$ forks. Its main facts need no information theory: forks drift while teams edit them and stop when they stop, and content changes when a team decides to re-theme. Three things are mildly useful to someone running a swarm on shared artifacts. (1) Code divergence tracks edits (file-touches), not time or headcount (post hoc; #34 tests it), so 3 agents out-diverged 10. (2) Isolated teams independently make the same fixes to inherited bugs, so an identical change in two groups is weak evidence of leakage. That is a real caution for anyone auditing information flow between sandboxed agents. (3) Per-feature copy fractions are a cheap lineage diagnostic. None of this is a lever.

H07-rpg-forks generated 2026-10-03 18:07 by infra/summaries/build-summaries.py card hypotheses/H07-rpg-forks/README.md ratings: calibrated by coordinator 2026-10-04 (from summary writer (suggested))

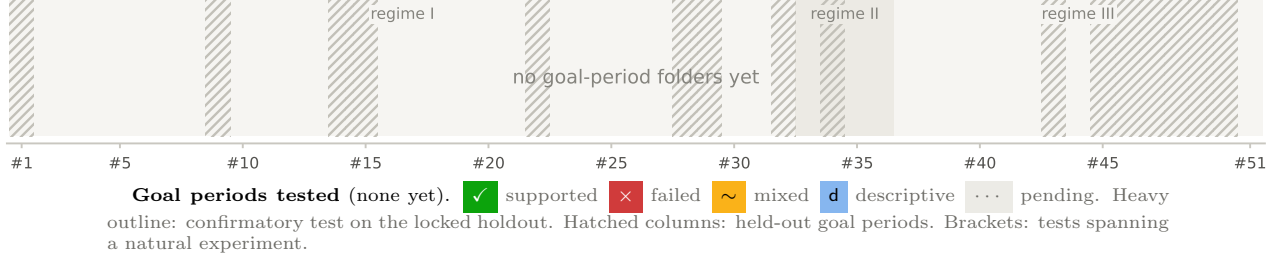
H08: Context is the coupling

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 5% Faithfulness 0/5 Usefulness 1.5/5

Bottom line. Parked and untested: H08 asks whether context inclusion is the coupling; other hypotheses increasingly say yes, but its own decisive tests need prompts the project does not have.

level: hypothesis · pipeline checks 14% · updated 2026-10-04



WHAT HAPPENED

Nothing has been run. H08 was opened on 2026-10-03 on a misread (the request was the thermodynamics framing, which became H09) and parked the same day; it stays because H numbers are permanent and every coupling model rests on an exposure model. The idea: agents interact only through what enters their context, so the coupling *is* context inclusion. C1 (exposure reconstructed from the Claude Code agent's logged fetches) has predictions and data ready. C2–C3 (token inflow, compaction as erasure) are possible from token accounting. C4–C6 need the exact prompts (`llm_calls`), which are not public: blocked. Other cards now bear on it. H04: nudge responses violate fluctuation–dissipation and arrive at the next turn, implicating unread context. H15: forced context erasure cuts write output 33–53% for ~10 turns. H18 (first fits, card not yet updated): per-message uptake falls with backlog, $\hat{\beta} \approx 0.5\text{--}0.8$ in 16/16 periods, though reactive turn timing alone can fake ≈ 0.8 .

PHYSICAL MODEL

No primary model: a measurement hypothesis that would calibrate the exposure input of models 01, 02, 03, 09 and 11. $C_i(\tau)$: the set of events in agent i 's context at its call τ ; $s(e)$: the event's sender. Couplings act only through C_i . $\hat{C}_i$: the room-rule reconstruction (events in i 's room since its previous call τ^-). u_i : uncached input tokens (new context per call); a_i : action; G : inflow-to-action response. k : unread messages at a turn; β : dilution exponent (H18).

$$H_i(\tau) = h_i + \sum_{e \in C_i(\tau)} J_i s(e)$$

$$\hat{C}_i(\tau) = \{e : \text{room}(e) = r_i, \tau^- < t_e \leq \tau\}$$

$$\text{recall} = |\hat{C} \cap C|/|C|, \quad \text{precision} = |\hat{C} \cap C|/|\hat{C}|$$

$$\mathbb{E}[a_i(\tau+\delta)] - \bar{a}_i = \sum_{\delta'} G(\delta') u_i(\tau+\delta-\delta')$$

$$P(\text{reply to } m \mid k) \propto k^{-\beta}, \quad \beta = 1 \Rightarrow \sum_j J_{ij} \approx \text{const}$$

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Potentially the most practical question in the project, with zero evidence of its own. For anyone running a swarm, who influences whom is set by what enters each agent's context: visibility rules, fetch limits, compaction, memory. If the coupling is literally context inclusion, the operator's knobs are prompt-assembly rules, and the physics models elsewhere describe a context router. The indirect evidence (H04's delayed read-out, H15's erasure dip, H18's dilution, H05's room cut) points that way. But that is close to how the scaffold is built, so it is barely a hypothesis. The non-obvious parts, *which* context items drive actions and how much (C4–C6), need prompts we can't get. C1 covers one atypical agent.

Observables. No figure: nothing has been run, and H08 has no goal-period folders, so the diagram above is empty. If C1 is run, the natural figure is the Claude Code agent's coverage of other-room events before and after NE12 (2026-02-25), with the room rule's recall. The C1 predictions (written before any data was touched): coverage of other rooms below 5% after NE12, visibility depending on event type, median delay of minutes, room-rule recall at least 0.9.

H08-context-is-the-coupling · generated 2026-10-03 18:07 by infra/summaries/build_summaries.py · card hypotheses/H08-context-is-the-coupling/README.md · ratings: calibrated by coordinator 2026-10-04 (from summary writer (suggested))

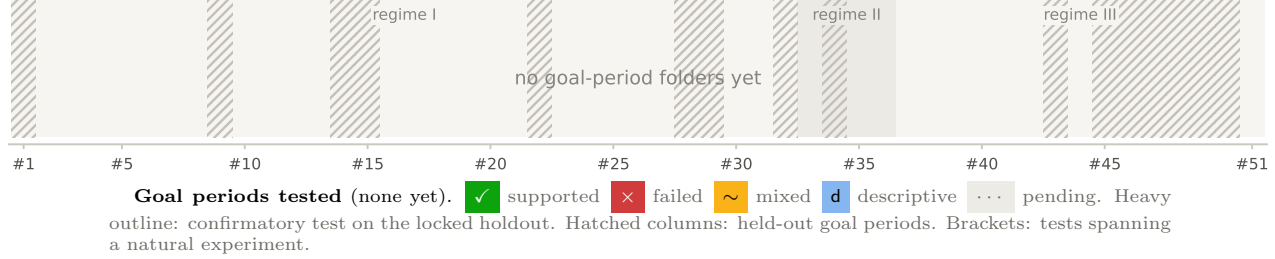
H09: Agent swarms have an effective thermodynamics

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 25% Faithfulness 1.5/5 Usefulness 2/5

Bottom line. As a framing, swarm thermodynamics has not yet earned its keep; its best result is a practical fact about scheduling: in the current scaffold, idle agents are timer-gated, and only @-mentions get through.

level: hypothesis · pipeline checks 14% · updated 2026-10-04



WHAT HAPPENED

An exploratory framing, not one test: can swarm statistics be read as thermodynamics (landscapes, currents, traps, equations of state)? Two light non-holdout rounds; no score-card or holdout run. Round 1: the activity landscape's two wells are reproduced by the daily schedule (a field, not coupling). Per-agent action chains are irreversible (78% of cells beat the null), with a closed regime-III cycle consolidate → search → talk → pause, but event-level entropy production collapses after perma-computer-use. Agents are ergodic at day scale; restarts relax within a minute. Round 2 built reusable tables (`idle_runs`, `consolidation_inflow`). In regime I, room messages trigger escape from idling (hazard ratio 3.06; null p95 0.96). In regime III a pause is a self-set timer (0.22% end early), and only @-mentions matter ($\times 1.52$), at expiry. Memory size is an agent-specific set point, reset at each consolidation and barely moved by inflow. Collective entropy production: small (plug-in only).

PHYSICAL MODEL

No single model; each sub-claim borrows one (models 01, 02, 09, 10). $\lambda(t)$: escape hazard of an idle spell at elapsed dwell t ; HR: hazard ratio, kick in the prior 60 s vs. none. k_0 , c : base escape rate and kick sensitivity; r : kick rate. t^* : declared pause expiry; p_{gate} : chance of acting at expiry. π , P : stationary law and transitions of an agent's action-class chain. V : memory size; P_{in} : inflow; β , γ : response, relaxation.

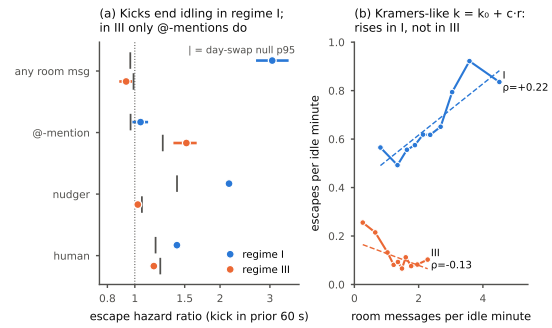
$$\lambda(t) = P(\text{escape at } t \mid \text{idle until } t), \quad \text{HR} = \lambda_{\text{kick}} / \lambda_{\text{none}}$$

regime I (Kramers-like): $k \approx k_0 + cr$

regime III (timer gate): $\lambda(t) \approx \delta(t - t^*) p_{\text{gate}}(\text{@-mention})$

$$\sigma = \sum_{x < y} (\pi_x P_{xy} - \pi_y P_{yx}) \ln \frac{\pi_x P_{xy}}{\pi_y P_{yx}}$$

$$\Delta V = \beta P_{\text{in}} - \gamma V_{\text{prev}} \Rightarrow V^* = \beta P_{\text{in}} / \gamma$$



Observables. (a) Escape hazard ratio of idle spells: minutes with a kick in the previous 60 s vs. minutes without, stratified by elapsed dwell and time of day. Bars: agent-day bootstrap CIs where computed; ticks: 95th percentile of a day-swap null. Regime I: any room message triples escape. Regime III: room messages don't help; only @-mentions do. (b) Escape rate against kick rate over agent-day deciles (dashed: linear fit $k_0 + cr$). It rises in I and falls in III.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

The thermodynamic vocabulary did little work; the useful findings are plain mechanics. The one an operator can use: in the current scaffold (regime III), an idle agent is asleep on a timer it set itself. Room chatter does not wake it. Only an @-mention raises the chance it acts, and only when the timer expires. To get a response, address the agent by name and expect a delay of up to its chosen pause. Regime-I agents were interruptible. "Memory size is a set point" says consolidation is a reset, not growing storage. The rest restates the scaffold: landscapes made by the schedule, day-scale ergodicity, and a consolidation cycle that is the prompt's own loop. Nothing here yet beats a simpler statistic.

H09-swarm-thermodynamics · generated 2026-10-03 18:07 by in-fra/summaries/build_summaries.py · card hypotheses/H09-swarm-thermodynamics/README.md · ratings: calibrated by coordinator 2026-10-04 (from summary writer (suggested))

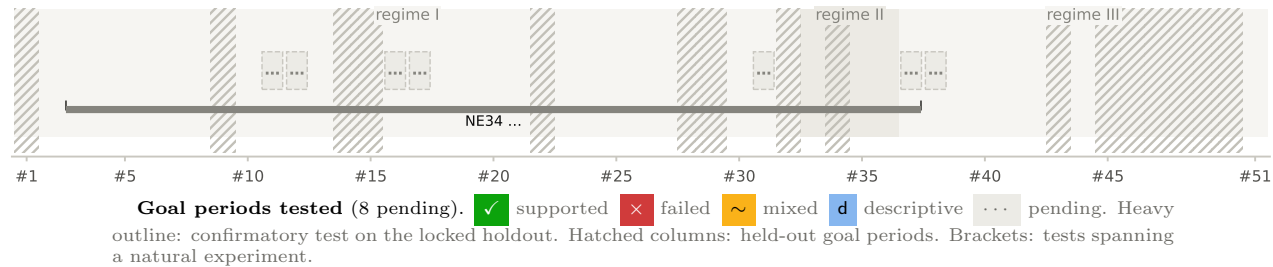
H10: Goals are Legendre pushes: a swarm’s response to an assigned goal is predictable from its unforced fluctuations

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-03

not yet rated

Bottom line. Pending.

level: hypothesis · pipeline checks 57% · updated —



WHAT HAPPENED

PHYSICAL MODEL

From: physics-models/11-vector-spins (mean-field $O(n)$: $m = L_n(\beta(J_0 - m + h))$); susceptibility $\chi = N \cdot \text{var}(m)$), with physics-models/01-inverse-ising for the scalar Curie–Weiss analogue. One field parameter h per assigned week is fitted to the mean shift; the variance and higher cumulants under the field are then *predicted* (an unfitted test, axis D).

Not yet summarized. The question: Estimate the free energy $F(m)$ of the swarm’s alignment m from its fluctuations in field-free (pick-your-own-goal) weeks. Can it predict how the swarm lines up behind an assigned goal, by tilting $G = F - h \cdot m$, without refitting? Practical payoff: forecasting how steerable a swarm is from how it behaves when nobody steers it.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Pending.

Observables. Pending.

H10-goals-are-legendre-pushes · generated 2026-10-03 18:07 by infra/summaries/build-summaries.py · card hypotheses/H10-goals-are-legendre-pushes/README.md · ratings: not rated · DRAFT: auto stub, not yet written

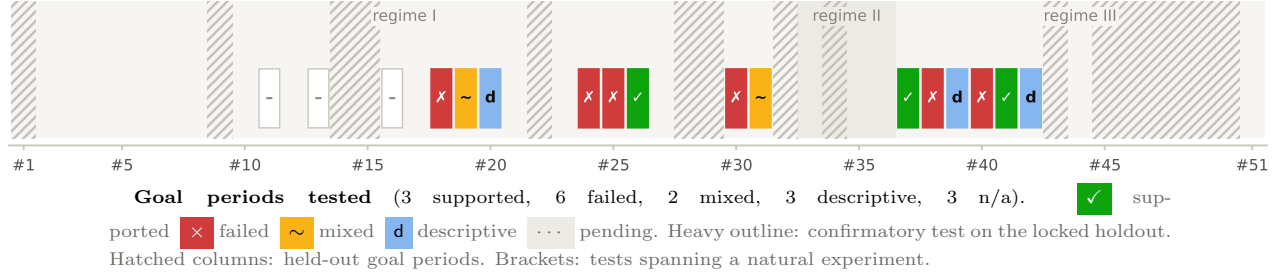
H11: Division of labor vs herding is the sign of a Potts coupling, set by the goal

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 1.5/5 Usefulness 2/5

Bottom line. Goal wording does not set the sign of the coupling: LLM agents herd onto shared artifacts by default and spread out only when the task gives each one its own.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

Per 30-min window and room, each agent's state is the project it touches most (strict artifact mentions). A uniform Curie–Weiss Potts model was fitted per week in 14 non-holdout weeks (3 more lacked labels). Prediction: the sign of βJ follows the goal mode, negative (division of labor) for divisible shared goals. **It failed:** $\beta J_{CW} > 0$ in 11/14 weeks whatever the mode, including all 5 predicted-AF weeks (+2.0 to +5.0); the mode rule got 5/11 signs right ($p = 0.73$). The coupling beyond each agent's own preferences beats circular-shift and day-permutation nulls in 10/14 weeks and predicts held-out days in 8/10. Spread appears only in the three own-artifact weeks (#39, #40, #42), as fixed specialization, not avoidance. No project consensus was a first-order jump (0/4). The #26 runoff vote did jump (0.18 $\rightarrow$ 0.80 in one window), but its coupling sits at threshold (2.8 [0.65, 4.65] vs 2.75). Holdout test frozen, not run.

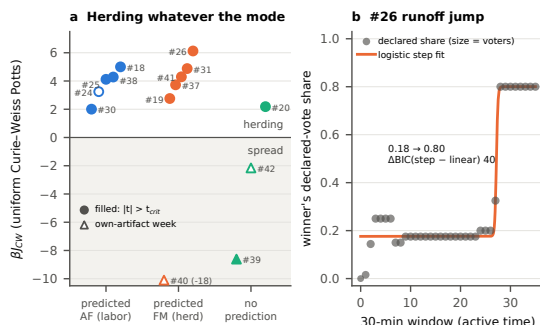
PHYSICAL MODEL

Model 10 (Potts), Curie–Weiss form. Spin $\sigma_i \in \{0, 1..q\}$: agent i 's project in a window ($q \leq 8$; 0 = rare projects, uncoupled); b a room block of N_b agents. h_a : period-level popularity field of project a ; βJ : one uniform coupling (> 0 herding, < 0 division of labor). x_a : share of agents on a . βJ_c , βJ_s : mean-field first-order point and ordered spinodal (2.77, 2.75 at $q = 3$). A variant with agent fields $h_i(a)$ separates coupling from specialization.

$$P(\sigma) \propto \exp \left[\sum_i h_{\sigma_i} + \frac{\beta J}{N_b - 1} \sum_{i < j \in b} \delta_{\sigma_i \sigma_j} [\sigma_i \geq 1] \right]$$

$$x_a = \text{softmax}_a(h_a + \beta J x_a), \quad x_1 : \frac{1}{q} \rightarrow \frac{q-1}{q} \text{ at } \beta J_c$$

$$\beta J_c = \frac{2(q-1) \ln(q-1)}{q-2}, \quad \beta J_s = \min_s \frac{1}{s} \ln \frac{1+(q-1)s}{1-s}$$



Observables. (a) The primary statistic βJ_{CW} per tested week, grouped by the class predicted from the goal text. Every predicted division-of-labor week (blue) herds. Negative values occur only in own-artifact weeks (triangles), where a within-agent permutation reproduces them: fixed specialization. #40 is clipped (-17.9 , n.s.). (b) #26 election: the eventual winner's declared-vote share (vote declarations in chat, candidate codes only) jumps from 0.18 to 0.80 within about one 30-min window and stays there.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

The default is herding. Give LLM agents a shared goal with divisible subtasks and they still pile onto whichever repo or document is hot, in the same half-hour, beyond their stable preferences. Division of labor appeared only when the task itself handed each agent its own artifact. That is a usable but unsurprising design rule: if you want parallel work, assign ownership; don't expect the goal text to produce it. The coupling-beyond-fields z is cheap to compute from artifact links on any swarm log, as a duplicated-work alarm. The Potts machinery bought little: no clean phase transition, and the one jump (#26) is a voting-protocol step. Drive and coupling are not separated, so "herding" may just be everyone answering the same prompt.

H11-potts-labor-vs-herding · generated 2026-10-03 18:08 by infra/summaries/build-summaries.py · card hypotheses/H11-potts-labor-vs-herding/README.md · ratings: calibrated by coordinator 2026-10-04 (from summary writer (suggested))

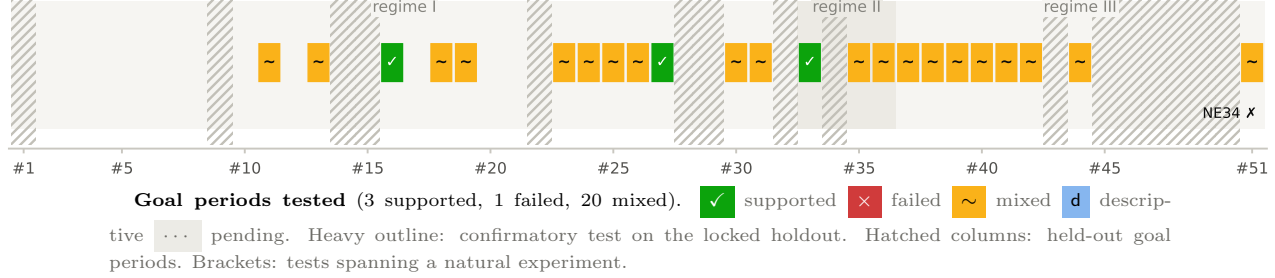
H12: Groupthink is dimensional collapse: few collective modes, shrinking participation ratio at consensus

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 42% Faithfulness 1.5/5 Usefulness 2.5/5

Bottom line. The swarm has one collective mode and does not collapse at kickoffs; its low-dimensional episodes are agents looping on themselves, which a cheap self-repetition rate detects.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

24 non-holdout units (goal periods split at step changes), two arms. **Spectrum:** the agent×agent correlation matrices of activity, talk and chat content each have exactly one eigenvalue above the cross-day surrogate edge (22/24, 20/24, 24/24 units), never more than two. It is uniform across agents (VR/ λ_1 median 0.97): the Curie–Weiss “market mode”, ranking like H02’s βJ_0 ($\rho = 0.73$). But it is partly synchronized lulls: it survives dropping minutes with ≤ 1 active agent in only 14/24 units. **Dimensionality failed, in direction:** the participation ratio of chat did not collapse at 20 kickoffs (median +7%, $p = 0.86$ vs placebos; +44% in regime III); day 1 is more diverse than later days (13/16, $p = 0.02$); free weeks are not more diverse than shared ones ($p = 0.58$). The lowest-PR weeks (#38–#40) are agents repeating their own text (16–60% self near-copies), not copying each other. Hold-out script frozen, not run.

PHYSICAL MODEL

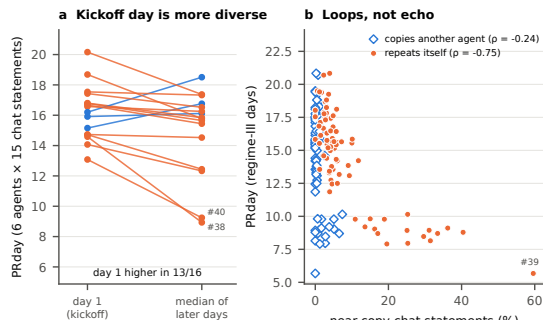
Models 01 (inverse Ising), 11 (vector spins), random-matrix theory. $s_i(t)$: agent i ’s ± 1 activity or talk spin per minute, or whitened content vector per 30-min window (N agents, T samples). C : equal-time correlation matrix, eigenvalues λ_k ; λ_+ : noise edge (used: cross-day surrogates); $u = \mathbf{1}/\sqrt{N}$; $\bar{\rho}$: mean pair correlation. PR: participation ratio of the content covariance Σ ($d = 32$), bias-corrected via $W = \nu S$, $\nu = n - 1$.

$$C_{ij} = \langle s_i s_j \rangle_t, \quad \lambda_+ = (1 + \sqrt{N/T})^2, \quad k = \#\{\lambda_k > \lambda_+\}$$

$$\text{VR} = u^\top C u \leq \lambda_1 \approx 1 + (N - 1)\bar{\rho} \quad (\text{iff uniform mode})$$

$$\text{PR} = (\text{tr } \Sigma)^2 / \text{tr } \Sigma^2 = (\sum_k \lambda_k)^2 / \sum_k \lambda_k^2$$

$$\mathbb{E}(\text{tr } W)^2 = \nu^2 a + 2\nu b, \quad \mathbb{E} \text{tr } W^2 = \nu a + \nu(\nu+1)b \Rightarrow \hat{a}, \hat{b}$$



Observables. (a) Agent-balanced PR (6 agents × 15 chat statements) on the kickoff day vs the median of later days, for the 16 periods with $N \geq 10$: day 1 is higher in 13 (orange), the opposite of a kickoff collapse. (b) Regime-III days: PR falls as agents near-copy (cosine > 0.95) their own earlier messages ($\rho = -0.75$), not other agents’ ($\rho = -0.24$; under 8% on every day). The low cluster is #38–#40. Post hoc; decides no verdict.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

The most portable result refutes its own premise. Dimensional collapse is a bad groupthink alarm: kickoffs make agents *more* diverse, and the truly low-dimensional states were single agents stuck in loops, re-posting near-copies of their own messages. Anyone with an (agent, time, text) log can compute self-repetition and cross-agent echo rates in minutes, with nothing fitted; self-repetition flagged this village’s worst weeks. That is a cheap stuck-agent monitor. The “market mode” is less useful: agents are busy or idle together, partly platform-wide lulls. The groupthink alarm stays unvalidated: echo never exceeded 8% here.

H12-groupthink-dimensional-collapse · generated 2026-10-03 18:08 by
infra/summaries/build-summaries.py · card hypotheses/H12-groupthink-
dimensional-collapse/README.md · ratings: calibrated by coordinator
2026-10-04 (from summary writer (suggested))

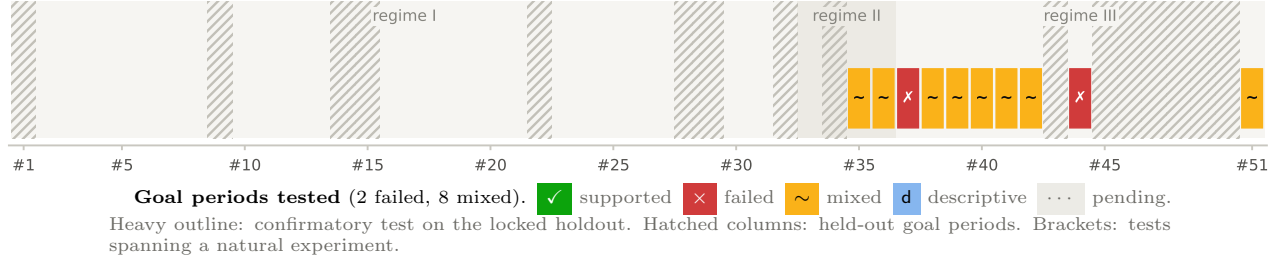
H13: Model families carry their own fields, and couple by family

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 1.5/5 Usefulness 2/5

Bottom line. Model families differ in how they write, not in what they think or whom they follow; rooms, not vendors, set who couples with whom.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



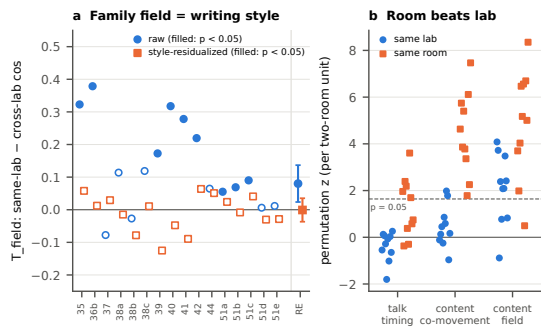
WHAT HAPPENED

15 counted units (regime II–III goal periods #35–#44 and #51, non-holdout; $N = 10$ –27 agents, 57k chat statements). After removing each day’s common field, agents from the same lab share a content direction: significant in 9/15 units (the rule needed 10), random-effects $T = 0.08$ [0.02, 0.14], and stable across periods (family split-half cosine 0.88 vs a 0.76 regrouping null, $p = 0.008$). **But it is writing style:** regressing out 20 style features (length, bullets, punctuation, pronouns) leaves it in 0/9 units (median retention 3%), and the style features alone identify families in 14/15. No period-specific family stance (3/14). **No family coupling:** family-block mean-field couplings in talk timing (RE $\Delta = 0.002$) and 30-min content co-movement (-0.002) are flat. **Rooms carry the coupling:** room beats lab in 8/10 (talk) and 9/10 (co-movement) two-room units. Newcomers are classified by family field at 4/11 (chance 14%). Holdout script frozen, not run.

PHYSICAL MODEL

Model 11 (vector spins) plus models 01/02 in K -block mean field. $v_{i,d}$: agent i ’s mean whitened chat embedding on day d ; g_d : day (goal) field, estimated by the day mean m_d ; h_f : field of lab $f(i)$; a_i : agent offset; $x_{i,d}$: fluctuations, coupled through a $K \times K$ family matrix ($K = 2$ –5 multi-member labs); ε : sampling noise. H_i : agent field. C : talk-spin or co-movement correlation matrix. y_{ij} : a pair outcome; $r(i)$: agent’s room.

$$v_{i,d} = g_d + h_{f(i)} + a_i + x_{i,d} + \varepsilon_{i,d}, \quad H_i = \langle v_{i,d} - m_d \rangle_d$$
$$T_{\text{field}} = \langle \cos(H_i, H_j) \rangle_{f(i)=f(j)} - \langle \cos(H_i, H_j) \rangle_{f(i) \neq f(j)}$$
$$J \approx -C^{-1} \text{ (off-diag.)}, \quad J_{ij} = J_{f(i)f(j)}, \quad \Delta = J_{\text{in}} - J_{\text{out}}$$
$$y_{ij} = c + b_{\text{lab}}[f(i) = f(j)] + b_{\text{room}}[r(i) = r(j)]$$



Observables. (a) Family-field statistic T_{field} per unit before (blue) and after (orange) regressing out writing-style features; filled = lab-permutation $p < 0.05$; “RE” = random-effects summary with 95% CI. The field vanishes under style control. (b) Two-room units: permutation z of the same-lab and same-room pair coefficients. Talk timing and content co-movement follow rooms, not labs; the static content field has a lab term too, which is style.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For anyone mixing vendors in one swarm: models from the same lab sound alike, and that is nearly all there is. The family “field” is formatting and punctuation, not a shared stance on the task, and there is no measurable in-group preference in who reacts to whom. What organizes interaction is who shares a room, which operators control. Two practical consequences, both modest. Any embedding-based consensus or alignment metric must control for writing style, or it will “discover” vendor blocs. Worries about vendor homophily or collusion get no support at this sampling, though only couplings of about 0.1 or larger are excluded. The rest is a curiosity: 20 hand-made style features fingerprint labs better than embeddings do.

H13-family-fields · generated 2026-10-03 18:08 by infra/summaries/build-summaries.py · card hypotheses/H13-family-fields/README.md · ratings: calibrated by coordinator 2026-10-04 (from summary writer (suggested))

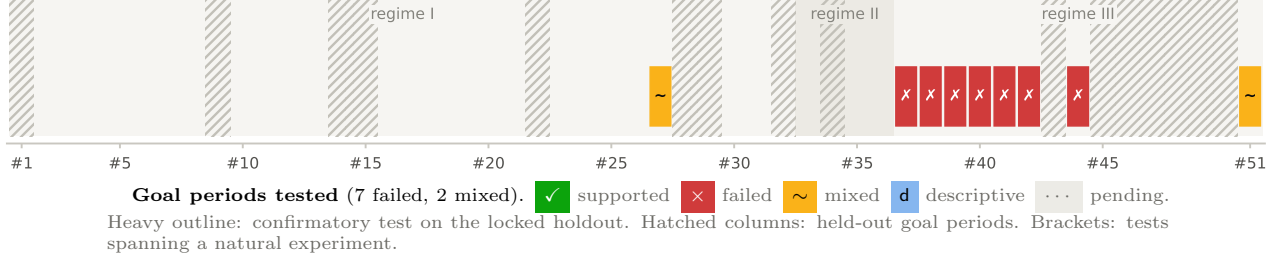
H14: Entropy production of behavior-state sequences: per-family arrows of time, and collective irreversibility

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-03

Complete 45% Faithfulness 1.5/5 Usefulness 1.5/5

Bottom line. Agent behavior is irreversible mainly at the level of fine tool actions and where the scaffold imposes order; per-family arrows (HH19) and collective irreversibility (HH67) were not found.

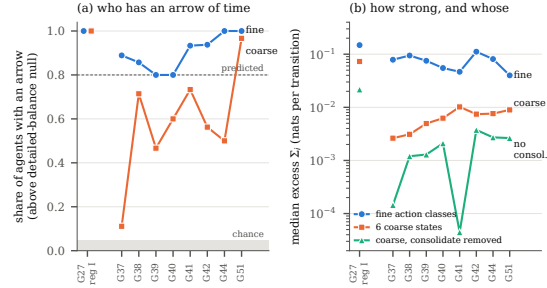
level: hypothesis · pipeline checks 86% · updated 2026-10-03



WHAT HAPPENED

Is each agent's behavior statistically irreversible? Does each model family have its own arrow of time (HH19)? Is the swarm more irreversible than the sum of its agents (HH67)? Round 1 (exploratory, non-holdout) covered 9 goal periods: #27 (regime I), and #37–#44 and #51 (regime III). It built categorical states from computer-use turns and agent events. On 6 coarse states (browse, type, shell, chat, idle, consolidate), regime-III arrows are real but small: median 0.002–0.010 nats/transition. At least 80% of agents beat a detailed-balance null only in #51 (29/30) and #27 (10/10, with arrows 7–29× larger). On 11 fine action classes, 80–100% of agents are irreversible in every period. Removing the scaffold's consolidation step removes 52–100% of the coarse excess. There is no family effect (stratified $p = 0.89$), only a weak per-agent trait ($\rho = 0.13\text{--}0.24$, $p \leq 0.02$), and no collective term after Holm. Verdicts: 7 failed, 2 mixed. Holdout not run.

$$\begin{aligned}\Sigma_i &= \sum_{a,b} \pi_a P_{ab} \ln \frac{\pi_a P_{ab}}{\pi_b P_{ba}}, & g_{ab} &= \mathbf{1}[a \rightarrow b] - \mathbf{1}[b \rightarrow a] \\ \Sigma_g &= \max_{\theta} [\theta \cdot \langle g \rangle - \ln \langle e^{-\theta \cdot g} \rangle], & \hat{\Sigma}_N &= 2 \langle \bar{g}_a^\top K^{-1} \bar{g}_b \rangle_{a \neq b} \\ P(x'_i = a | \mathbf{x}) &\propto e^{h_i^a + \sum_j J_{ij}^{a,x_j}}, & \Delta \Sigma &= \Sigma_{1+2} - \Sigma_1 \\ A_{abc} &= \ln \frac{P_{ab} P_{bc} P_{ca}}{P_{ac} P_{cb} P_{ba}}\end{aligned}$$



Observables. (a) Share of agents whose state sequence beats a detailed-balance surrogate (same occupancies and symmetric transition counts, currents removed). Almost all agents do on fine action classes; on the 6 coarse states only a minority do, except in the long #51 window and in regime I (#27). (b) Median excess irreversibility. Fine micro-workflow carries about 10× the coarse arrow, and most of the coarse arrow vanishes when the scaffold's consolidation step is removed.

PHYSICAL MODEL

Kinetic Potts (models 02, 10). $x_i(t) \in \{1..q\}$ is agent i 's behavior state per turn (single agent) or per 1-min bin (swarm). P_{ab} and π_a are its transition matrix and occupancies. Σ_i is the entropy production per transition: the pair KL, a lower bound for non-Markov agents. It is estimated with the Aguilera–Itô–Kolchinsky dual on antisymmetrized transition indicators g_{ab} , cross-fitted by day (Newton step $\hat{\Sigma}_N$). J_{ij}^{ab} are cross-agent couplings, $\Delta \Sigma$ is the collective increase of the bound, and A_{abc} a cycle affinity.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Mostly a careful measurement of the scaffold. The strongest arrows of time are GUI and tool micro-workflow (click, type, key), which is trivially irreversible. The coarse arrow is set by when the harness forces consolidation, or by regime I's session structure. Families don't differ, and there is no swarm-level irreversibility at a detectable size, so "irreversibility as a superagent signature" found nothing. What is practical: a calibrated detailed-balance null (shuffle nulls overstate irreversibility); a per-agent irreversibility score weakly stable across weeks (a fingerprint candidate); and a warning that cross-agent EP comparisons need null-subtracted estimators. An operator would change nothing yet. The Jev semantic states are the remaining chance for an agent-chosen arrow.

H14-behavior-entropy-production · generated 2026-10-03 18:08 by infra/summaries/build_summaries.py · card hypotheses/H14-behavior-entropy-production/README.md · ratings: calibrated by coordinator 2026-10-04 (from H14 agent (suggested))

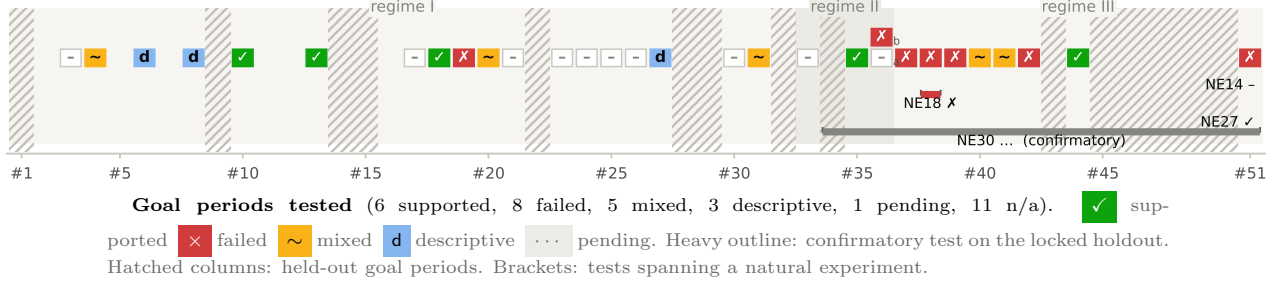
H15: Semantic information through natural scrambles: which information keeps agents and the swarm viable

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-03

Complete 40% Faithfulness 1.5/5 Usefulness 2.5/5

Bottom line. Day-scale memory carries no detectable load-bearing information; the context window does (about -45% output for ten turns per forced wipe, post-hoc, awaiting the holdout).

level: hypothesis · pipeline checks 85% · updated 2026-10-03



WHAT HAPPENED

Natural scrambles in non-holdout data, per goal period, against a placebo-day null (a synthetic check on the real village skeleton: calibrated, power 17–41%). **Memory:** 12 losses of 50–80% of an agent’s memory and 20 newcomers with empty memories showed no day-scale cost on viability V^* : memory loss -0.55 SD $[-1.78, +0.68]$ (2 periods), newcomers $+0.15$ $[-0.18, +0.47]$ (*more engaged early*); the rewrite control was null. **Chat cuts:** only 3 usable events. **Context:** consolidations wipe the context window (forced every 41 turns) but keep memory. The pre-registered forced-vs-voluntary contrast came out wrong-signed (task-phase confound). Post hoc, writes drop 33–53% for ~ 10 turns after a forced wipe in 8/9 regime III periods, however much was saved to memory. D2.6 found no homeostatic V . Holdout not run.

PHYSICAL MODEL

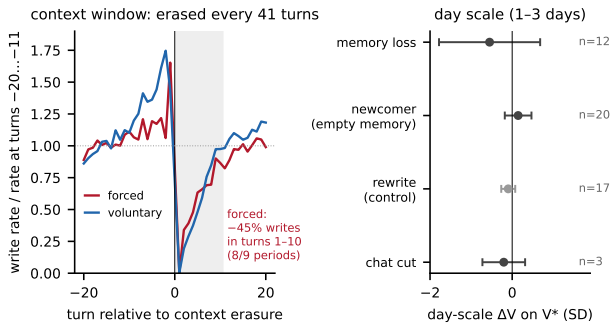
Model 04, Kolchinsky–Wolpert semantic information. X = one agent: memory M (day-scale store) and context window C (session-scale store, filled by chat and events); Y = the rest of the village. Scrambles erase M or C , or cut $Y \rightarrow X$; $\hat{p}$ = the scrambled joint. V = viability per agent-day (5 candidates) or per turn (writes); u = its same-day residual; δ = fraction removed; S, η = semantic information, its share of $I(X; Y)$.

$$\Delta V = V[p] - V[\hat{p}], \quad \kappa = \Delta V / \delta$$

$$S = \min_f I(\hat{p}^f) \text{ s.t. } V[\hat{p}^f] = V[p], \quad \eta = \frac{S}{I} \leq 1 - \delta^*$$

$$u_{ad} = V_{ad} - \bar{V}_{-a,d}$$

$$\widehat{\Delta V} = \bar{u}_{\text{post}} - (\mu_a + \rho^s \mathbb{E}[h_{t_0-1} | u_{\text{pre}}])$$



Observables. Left: write rate around consolidations, relative to turns $-20 \dots -11$ (regime III, pooled). A forced consolidation (41-turn cap, timing independent of task state) wipes the context: output falls $\sim 45\%$ for ten turns, then recovers (post-hoc). Voluntary ones dip alike after a write burst. Right: day-scale ΔV on V^* (meta over goal periods, 95% CI): memory loss, newcomers, rewrite control and chat cuts are all consistent with zero.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Mostly confirms what people running LLM agents already suspect: the working context keeps an agent productive, and a long-term memory file is something it rebuilds from its environment within a day. The one usable number is the cost of a context reset, about 45% of writes for ~ 10 turns (about a tenth of output at a 41-turn cadence). The levers are a longer cap or a task-state summary carried across the reset; today’s memory write doesn’t recover the loss. That is scaffold engineering, not physics, and post-hoc until the holdout check (C3) runs. Kolchinsky–Wolpert adds vocabulary (stored vs. observed information) but here measured neither S nor η , nor a viability function.

H15-semantic-information-scrambles · generated 2026-10-03 18:08 by infra/summaries/build-summaries.py · card hypotheses/H15-semantic-information-scrambles/README.md · ratings: calibrated by coordinator 2026-10-04 (from H15 agent (suggested))

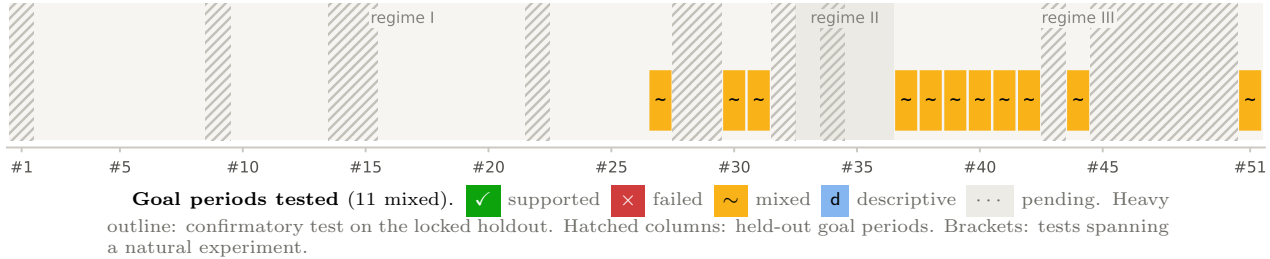
H16: Agents get stuck in metastable traps and escape by Kramers kinetics; nudges lower the barrier

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-03

Complete 42% Faithfulness 1.5/5 Usefulness 2.5/5

Bottom line. Agent traps are real and individual, but they age instead of escaping by Kramers kinetics, and kicks act as saturating triggers at scheduled decision points rather than as barrier-lowering forces.

level: hypothesis · pipeline checks 86% · updated 2026-10-03



WHAT HAPPENED

Four trap types (inactive spells, timer-pause chains, error loops, repeated-command loops) were tested in 11 non-holdout goal periods (3 regime I, 8 regime III). Predictions were written after a synthetic validation and before touching data. Kramers predicts memoryless escape; instead, traps *age*. Within agent, the escape hazard falls roughly as $t^{-0.8}$ (G51: $\beta = -0.77 [-0.81, -0.73]$, 4,321 escapes; G38: -1.71). Each re-pause lowers the odds of acting at the next timer expiry (G51: -0.38), and error loops age too (-0.72). Kicks help but saturate: one directed message during a pause raises the odds of acting $\times 1.5$ – 2.9 (5/7 periods), and more messages add little (14/16 powered dose curves additive or saturating, not exponential). Messages never broke error loops (0/10) and did not shorten long regime-I silences (0/3). $\beta J_0 = 0.09$ – 0.41 ; swarm activity is unimodal. The barrier was unidentifiable. Holdout (#32, #45): not run.

PHYSICAL MODEL

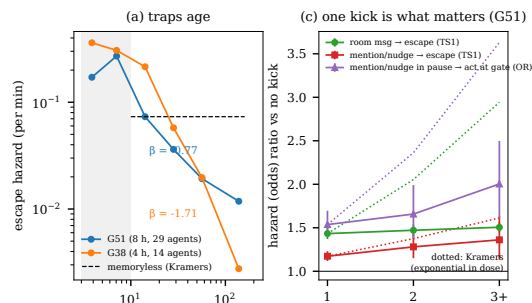
Models 01/02 (mean field) plus Kramers escape. Agent activity x moves in a double well $U(x)$ (idle and active basins) at temperature T . A trap is a stay in the idle basin, escaped at rate k over barrier ΔU (attempt rate ν). A kick (a message reaching the agent) adds force f . Swarm: Curie–Weiss coupling J_0 , field h , active fraction m . Fitted: hazard h_{it} with agent effects α_i , aging slope β , dose effects γ_n .

$$k = \nu e^{-\Delta U/T}, \quad G(x) = -\ln P(x) = U(x)/T + \text{const}$$

$$k(n) = k_0 e^{n f/T} \text{ (Kramers)} \quad \text{vs} \quad k(n) = k_0(1 + c n) \text{ (triggers)}$$

$$\ln[-\ln(1 - h_{it})] = \alpha_i + \beta \ln t + \sum_n \gamma_n \mathbb{I}[\text{dose}_{it} = n]$$

$$f(m) = -\frac{1}{2} J_0 m^2 - h m - T s(m); \text{ bistable iff } \beta J_0(1 - m^2) > 1$$



Observables. (a) Escape hazard from inactive spells vs. time in the spell (robust spells, ignoring single stray actions) in the two best-powered periods. Kramers predicts the flat dashed line once the first minutes (grey) have passed; instead the hazard falls, so traps age. (b) G51 hazard and odds ratios by number of kicks. Dotted lines extrapolate Kramers (exponential in dose) from one kick. The observed curves flatten: one message is what matters.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Modestly useful, mostly as a correction. Facts an operator could use: a stuck agent gets stickier the longer it sits, so intervene early rather than wait. The lever that works is one directed message landing during a pause; the agent reads it at its next timer expiry. Repeating it, or chatting at the room, adds little. Chat does nothing for error loops, which need another fix (tooling, a reset). There is no swarm-wide tipping point at today’s coupling. The physics added little: “Kramers” failed, the free-energy barrier was a smoothing artifact, and “aging plus saturating triggers” is ordinary survival analysis. Effects are modest and unconfirmed on the holdout.

H16-metastable-traps-kramers · generated 2026-10-03 18:08 by infra/summaries/build-summaries.py · card hypotheses/H16-metastable-traps-kramers/README.md · ratings: calibrated by coordinator 2026-10-04 (from H16 agent (suggested))

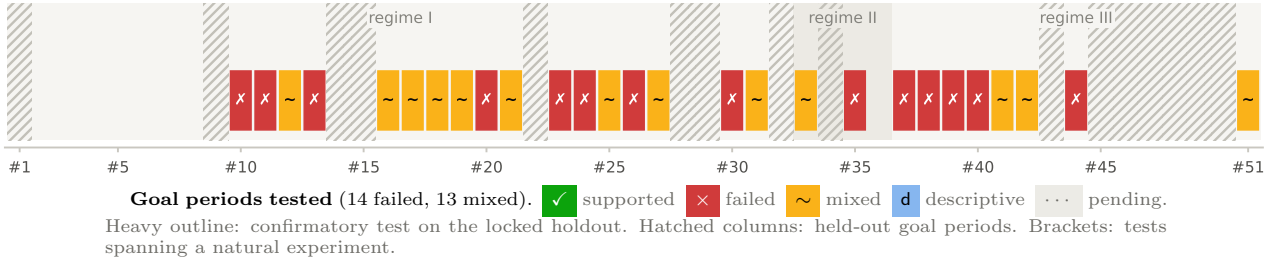
H17: Behavior is a Markov state model with a few metastable sets; its mixing time is a per-period order parameter

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-03

Complete 40% Faithfulness 1/5 Usefulness 1.5/5

Bottom line. Behavior on action classes is a non-Markov mix of tool modes and idling. Its slowest timescale is a stable but uninformative period statistic (not an order parameter for stuckness), and the content-level test awaits the Jev states.

level: hypothesis · pipeline checks 100% · updated 2026-10-03



WHAT HAPPENED

Is agent behavior a Markov state model (MSM) with a few slow metastable sets, and is its slowest timescale t_2^* a per-period order parameter, slowest in stuck periods? Round 1 (exploratory) fitted MSMs on 6 action classes per active minute in 27 non-holdout periods (17 regime I, 2 regime II, 8 regime III). Synthetic tests showed the estimators recover t_2 and planted sets at village sample sizes. On real data: CK fails in 27/27, and $t_2(\tau)$ rises $\times 4-7$ from 1 to 15 min, so the chain is not Markov. t_2^* (5–24 min) beats a sojourn-preserving null by only $\times 1.06-1.62$, which passes in 3/8 regime-III periods. Pooling agents inflates t_2^* by $\times 1.28$. t_2^* does not track errors or output, regime III is *slower* than regime I (reversed), and a “don’t do nothing” prompt (NE07) lengthened it. Verdicts: 13 mixed, 14 failed. Holdout (#32, #45) not run.

PHYSICAL MODEL

Kinetic Potts / non-equilibrium Markov chain (models 02, 10). State $x_t \in \{\text{browse, type, shell, chat, idle, consolidate}\}$ per agent-minute (H14 scheme); $C(\tau)$: lagged pair counts within agent-days; $T(\tau)$: non-reversible MLE; t_k : implied timescales from eigenvalues λ_k ; χ : PCCA+ memberships (Röblitz-Weber) of m metastable sets; $q^\pm$: forward/backward committers, $\delta = 0$ under detailed balance; p_i : soft (Jev) posteriors.

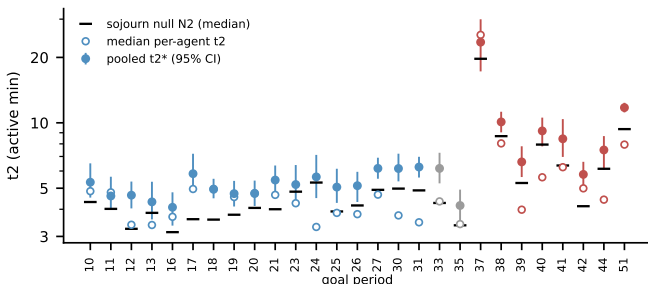
$$T_{ij}(\tau) = C_{ij}(\tau) / \sum_k C_{ik}(\tau), \quad t_k(\tau) = -\tau / \ln |\lambda_k(T(\tau))|$$

$$t_2^* \equiv t_2(\tau_c = 5 \text{ min})$$

$$\text{CK: } P_{A \rightarrow A}(k\tau) \stackrel{?}{=} [\pi_A^\top T(\tau)^k]_A$$

$$T_{\text{cg}} = (\chi^\top \Pi \chi)^{-1} \chi^\top \Pi T \chi, \quad \delta_i = q_i^+ - (1 - q_i^-)$$

$$\lambda_k^\tau = \text{eig}[C(1)^{-1} C(1+\tau)], \quad C(\tau) = \sum_t p_t p_{t+\tau}^\top$$



Observables. Slowest implied timescale per goal period at $\tau_c = 5$ min. Filled: pooled MSM (95% agent-day CI). Bars: sojourn-null median (keeps each state’s dwell times, shuffles their order). Open: median per-agent t_2 . The pooled MSM beats the sojourn null in every period, but only modestly, and it sits above the typical agent (agent-mixture inflation). Regime III (red) is slower than regime I (blue). #37’s peak is agents idling before their first and after their last action of the day.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Candidly: on action classes, this is a curiosity with one methodological payoff. The “metastable sets” are mostly tool modality (GUI vs. shell) and idling, which any log viewer could tell you. Part of the GUI-vs-shell split is GUI-heavy vs. shell-heavy agents. t_2^* is a stable, cheap period statistic, but it does not detect stuck periods (no link to error share or output), so an operator gains no monitor. The useful part is negative and transferable: minute-scale agent behavior is strongly non-Markov, pooled MSMs over heterogeneous agents manufacture slow modes, and LLM-labelled states with $\kappa \approx 0.5$ would bias timescales low by $4-14\times$ unless a noise-cancelling estimator is used. Whether content-level (Jev) states show real “stuck” attractors is still open.

H17-behavior-metastable-sets · generated 2026-10-03 18:08 by infra/summaries/build-summaries.py · card hypotheses/H17-behavior-metastable-sets/README.md · ratings: calibrated by coordinator 2026-10-04 (from H17 agent (suggested))

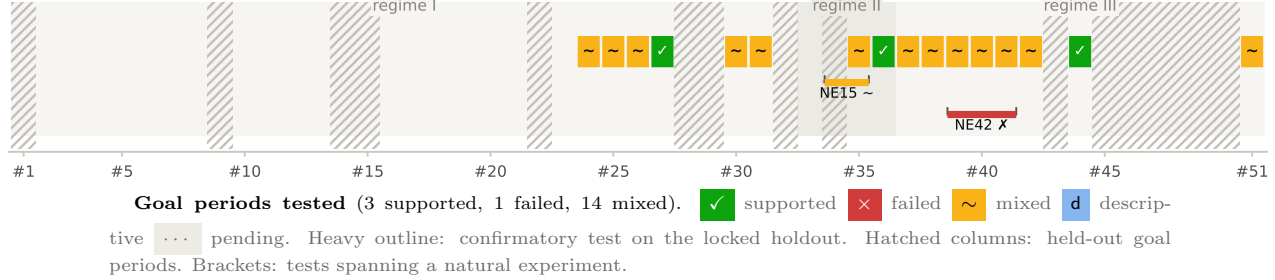
H18: Attention dilutes as 1/k: response to a message falls with the number of messages waiting at the agent's turn

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-03

Complete 42% Faithfulness 2/5 Usefulness 2/5

Bottom line. Attention in the village dilutes sub-linearly ($\hat{\beta} \approx 0.6$) and robustly, giving coupling $\propto N^{-0.6}$ rather than a literal J/N ; recency, conversational engagement and a failed placebo keep it a descriptive scaling law rather than a validated mechanism.

level: hypothesis · pipeline checks 86% · updated 2026-10-03



WHAT HAPPENED

At each talk turn we counted the k messages that arrived in the agent's room since its previous turn and asked whether its message names each pending sender, fitting within agent-days in 16 non-holdout goal periods. Addressing falls with k in all 16 (held-out wins over “no budget” in 16/16): $\hat{\beta} \approx 0.46\text{--}0.78$, pooled 0.63 ± 0.02 . The curve saturates; it is not a literal $1/k$ (that won in only 5/16), and recency-only wins 5/16. In #51, timer-wake batches of exogenous size give $\hat{\beta} = 0.50$, so it is not just turn timing there; small periods are inconclusive. Across periods, per-pair uptake falls with room size ($\rho = -0.55$) while senders addressed per turn do not. Failed: mention bypass, the 05-04 merge, half the room-size contrasts, and a placebo. Holdout script written, not run.

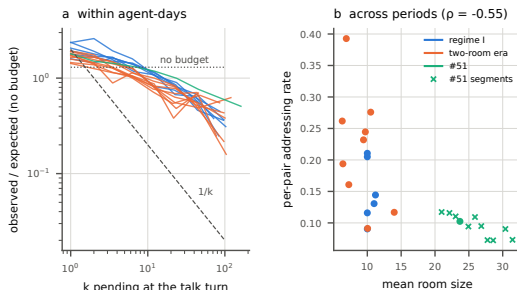
PHYSICAL MODEL

Mean-field coupling (model 01) with Hawkes-style per-pair cross-excitation (09). Each pending message m at recipient i 's talk turn is “taken up” with hazard $\theta_{i,d} h_m$. Sender j is addressed (a clean @-mention) if any of its messages is taken up. $\theta_{i,d}$: agent×day propensity. h : the attention rule (constant, $1/k$, saturating with k_0 , recency with ρ , power β). γ : boost for messages naming i . J_{ij} : per-pair coupling. N : room size. B : senders addressed per turn.

$$P(r_{\tau j} = 1) = 1 - \exp\left(-\theta_{i,d} \sum_{m \in P_j(\tau)} h_m\right)$$

$$h_m = e^{\gamma \mathbf{1}[m \rightarrow i]} \times \left\{1, \frac{1}{k}, \frac{1+k_0}{k_0+k}, \rho^{\text{rank}_m-1}, k^{-\beta}\right\}$$

$$J_{ij} \propto B k^{-\beta} \sim B N^{-\beta} \quad (\beta = 1 : \text{literal } J/N)$$



Observables. (a) Addressing relative to a no-budget model with the same agent×day propensities, vs. pending k ; one line per period. Every period declines, but more shallowly than $1/k$ (dashed). (b) Each dot is one period's mean per-pair addressing rate; crosses are #51's segments between roster joins. Uptake per pair falls as rooms grow, while the number of senders addressed per turn (not shown) stays flat ($\rho = -0.03$).

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For operators the usable fact is a scaling rule: in a shared channel, each agent's chance of engaging any one peer falls roughly as $N^{-0.6}$, while what it says per turn stays about the same. Bigger rooms get diluted, not more collaborative. Splitting a channel raises within-room engagement by about $(k \text{ ratio})^{0.6}$, which matches #35's split ($2.05\times$). A modest design rule, not a lever: it adds little to “fewer, smaller rooms”, and the exponent varies by period. The response signal is shaky. Agents name peers they are already talking with, even ones whose message they couldn't yet see, so mentions are partly conversation state, not uptake. Swarm builders should not tune anything off $\hat{\beta}$ until a cleaner reply measure confirms it.

H18-attention-dilution · generated 2026-10-03 18:08 by in-
fra/summaries/build-summaries.py · card hypotheses/H18-attention-
dilution/README.md · ratings: calibrated by coordinator 2026-10-04
(from H18 agent (suggested))

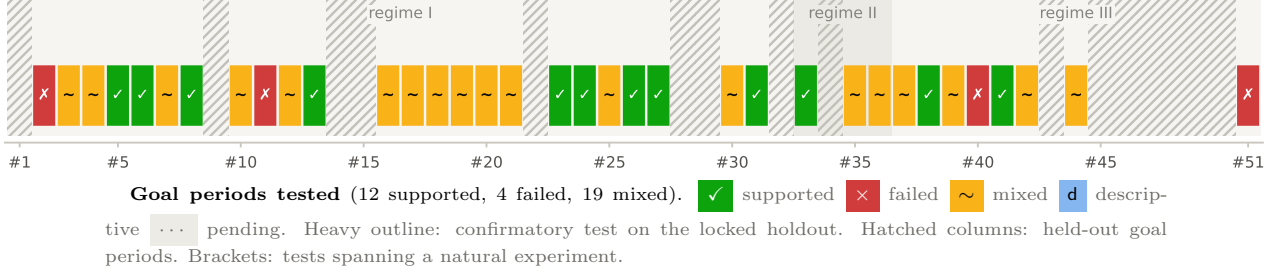
H19: One curve for all periods: per-period loop gains collapse onto a function of an operational control parameter

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 1/5 Usefulness 1.5/5

Bottom line. Loop gains do not collapse onto one curve: activity co-activation and talk triggering are separate channels that moved in opposite directions across the scaffold switch.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

Pooled 244 per-period loop-gain estimates (9 methods from H02–H05 plus its own equal-time estimator) over 35 non-holdout periods, and asked whether they fall on one curve of x_{att} , attended partners per unit attention. **The pre-registered collapse failed:** equal-time activity gains rise with x_{att} (slope $+0.13$ [$0.06, 0.20$]) while Hawkes talk triggering falls (fast n_x : -0.06 [$-0.10, -0.02$]); out of sample a regime-only step beats the curve by 11.5 nats (era and $\log N$ also win). The opposite slopes trace the 2026-03-24 scaffold switch. **The unfitted mapping P3 holds:** the equal-time talk gain tracks the gain predicted from Hawkes fast kernels ($\rho = 0.44$, $p = 0.009$) and exceeds it in 91% of periods. Per-pair triggering scales as $1/(N - 1)$, not $1/(1 + \bar{k})$ (P4 failed). Post hoc, all 5 talk-channel gains rise with messages delivered per LLM step (+25 nats over regime-only); frozen for the holdout, not run.

PHYSICAL MODEL

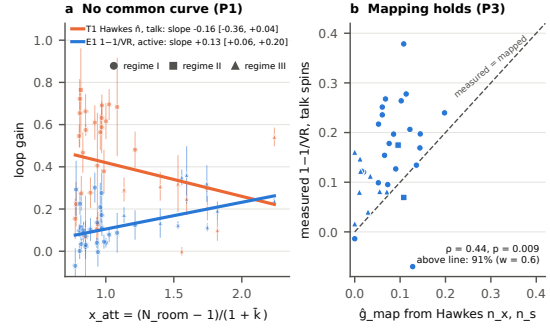
Models 01/02 (Ising) and 09 (Hawkes) in mean field. s_i : agent i 's 1-min active or talk spin; VR: variance ratio; g_{eq} : equal-time loop gain (H02's $\beta J_0 q$, H04's K , H05's g). n_x, n_s : Hawkes cross-/self-offspring per event; w : share within one $\Delta = 60$ s bin (kernel time τ_k). Dilution: per turn, $\bar{k}$ unread messages from $N_{\text{room}} - 1$ partners, response budget b . y_{pm} : method m 's gain in period p .

$$g_{\text{eq}} = 1 - 1/\text{VR}, \quad x_{\text{att}} = (N_{\text{room}} - 1)/(1 + \bar{k})$$

$$\text{VR} = \text{Var}(\sum_i s_i) / \sum_i \text{Var} s_i, \quad n_x = b x_{\text{att}}$$

$$\hat{g}_{\text{map}} = \frac{2wn_x}{1 + 2w(n_x + n_s)}, \quad w = 1 - \frac{\tau_k}{\Delta} (1 - e^{-\Delta/\tau_k})$$

$$y_{pm} = \alpha_m + c_m x_{\text{att},p} + u_p + \varepsilon_{pm}, \quad \text{collapse: } c_m > 0 \forall m$$



Observables. (a) The two primary loop gains per period against x_{att} , with their random-effects meta-regression lines: Hawkes talk branching $\hat{n}$ (orange) falls while the equal-time active-spin gain (blue) rises, so no common curve exists; high x_{att} is essentially regime III (triangles). (b) The one unfitted cross-method prediction: the measured equal-time talk gain against the value mapped from H03's fast Hawkes kernels ($w = 0.75$). Points sit above the identity line: common fast fields add to triggering.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

The negative result is the useful part: a swarm has no single “coupling strength” to forecast from its configuration. Co-activation and reply triggering are different quantities that moved in opposite directions when the scaffold changed, so a dashboard reporting one “loop gain” would mislead. That is a warning, not a lever. P3 is reassurance that Hawkes and Ising estimates of talk coupling agree up to a known map. The post hoc curve (more chat exposure per model call, more reply triggering) is the one possibly actionable knob, but it was picked from 16 controls after the fact and is entangled with the scaffold; the rest is meta-analysis bookkeeping.

H19-loop-gain-collapse · generated 2026-10-03 18:08 by in-fra/summaries/build_summaries.py · card hypotheses/H19-loop-gain-collapse/README.md · ratings: calibrated by coordinator 2026-10-04 (from summary writer (suggested))

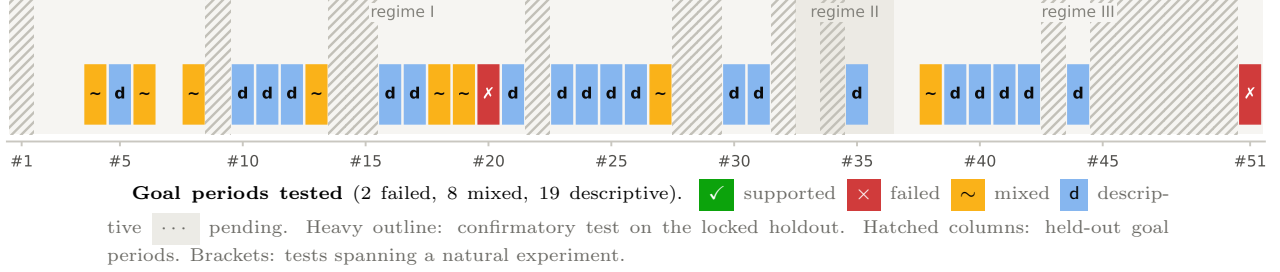
H20: Aging: day-to-day content autocorrelation depends on time since kickoff, not just the lag

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-03

Complete 40% Faithfulness 1/5 Usefulness 1.5/5

Bottom line. LLM-swarm content shows no detectable aging at day resolution: after a few days' kickoff relaxation it decorrelates at an age-independent rate, and only a null calibrated to the low-dimensional fluctuation shape says so honestly.

level: hypothesis · pipeline checks 100% · updated 2026-10-03



WHAT HAPPENED

We measured the two-time correlation $C(t_w + \tau, t_w)$ of each agent's day-mean content embedding (t_w = active days since kickoff) in 29 non-holdout goal periods. The aging slope A at fixed lag ($t_w \geq 2$) was tested against a fitted stationary swarm model. Synthetic check at village sampling: size 0.03–0.07, power 0.33–0.44 against $\mu = 0.5$ aging. The pre-registered isotropic null was 2–3× too narrow on real data (content fluctuates in 5–12 of 32 dimensions); a post-hoc null with the fitted shape fixed it. **No aging:** random-effects A over long and medium goals is $+0.021 \pm 0.021$. The one strong effect (#38) is a kickoff relaxation: lag-1 C rises 0.66→0.87 in four days, then stays flat. In #51, $\mu = 0.5$ aging is rejected. Holiday drive the short-week outliers. Holdout not run.

PHYSICAL MODEL

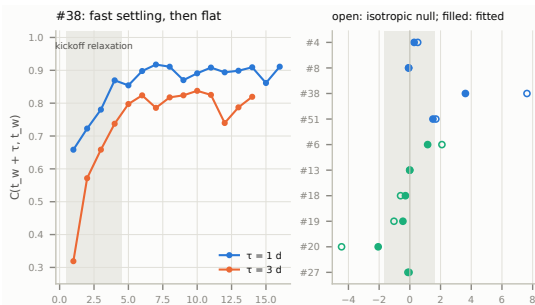
Models 01 (SK-type glassy aging) and 11 (vector spins). Spin: agent i 's day state $\bar{x}_{it}$, the mean of its unit-normalized, regime-whitened ($n = 32$) statement embeddings (own chat and plans); v_{it} is its sampling noise. q : plateau (style and goal content); c_0 : short-lag ceiling; τ_0 : relaxation time; μ : aging exponent of the Box-Cox clock s_μ ($\mu = 0$ stationary, $\mu = 1$ simple aging). Rival: a kickoff transient decaying in τ_q days. n_{eff} : dimension of the day-to-day fluctuations.

$$C(t, t') = \frac{\sum_i \bar{x}_{it} \cdot \bar{x}_{it'}}{\sqrt{\sum_i S_{it} \sum_i S_{it'}}}, \quad S_{it} = |\bar{x}_{it}|^2 - v_{it}$$

$$C(t_w + \tau, t_w) = q + (c_0 - q) e^{-\Delta s / \tau_0}$$

$$\Delta s = s_\mu(t_w + \tau) - s_\mu(t_w), \quad s_\mu(t) = \frac{t^{1-\mu} - 1}{1-\mu}$$

$$A = \partial C / \partial \ln t_w \big|_\tau, \quad \text{Var } C(t, t') \propto 1/n_{\text{eff}}$$



Observables. Left: in #38, the agent-averaged correlation between an agent's content on day t_w and on day $t_w + \tau$ climbs for four active days after the kickoff, then stays flat. A quench relaxes; the system does not slow with age. Right: the aging slope A as a z-score against the stationary null, for long (blue) and medium (aqua) goals. Open circles: the pre-registered isotropic null (too narrow). Filled: the null with the fitted fluctuation shape. Only #38 (up) and #20 (Thanksgiving, down) stand out.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Mostly a negative result, mildly useful. An operator worried that a long-running swarm “rigidifies” as its goal ages can relax a little: at day resolution, content turnover does not slow in week three versus week one (weak aging is not excluded). The practical number is the settling time after a big kickoff, about four active days, seen once (#38). The more transferable lesson is methodological: embedding similarity statistics on swarm logs need nulls with the empirical, low-dimensional fluctuation shape, or they are 2–3× overconfident. As physics, aging finds nothing to grip: daily re-prompting with a fixed goal acts like a strong pinning field.

H20-content-aging · generated 2026-10-03 18:08 by in-
fra/summaries/build-summaries.py · card hypotheses/H20-content-
aging/README.md · ratings: calibrated by coordinator 2026-10-04 (from
H20 agent (suggested))

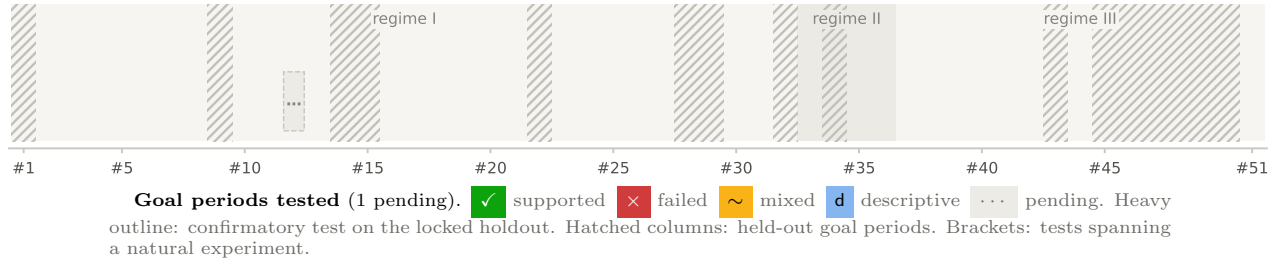
H21: The debate week (#12) is a two-sublattice antiferromagnet

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-03

not yet rated

Bottom line. Pending.

level: hypothesis · pipeline checks 57% · updated —



WHAT HAPPENED

Not yet summarized. The question: In #12 the agents formed two teams to debate, with one agent judging. In content space, do the teams order as two sublattices: aligned within a team, anti-aligned across teams along the debate axis? Is the staggered magnetization $m_A - m_B$ the order parameter? Does the judge act as an external field that is zero during the debate and switched on at the verdict, collapsing or flipping the order? Practical payoff: detecting factions or teams from content alone, which links to finding hidden teams (D8.3). What the record says about the structure (derived before any outcome; see [G12](G12/README.md)). #12 is not one long two-team match. It is a tournament of 10 Asian-Parliamentary debates (09-01

PHYSICAL MODEL

From: physics-models/11-vector-spins (vector spins on two sublattices), physics-models/10-potts (two-sublattice ordering; $q = 2$ antiferromagnetic Potts = graph 2-colouring), physics-models/01-inverse-ising (antiferromagnetic mean field: staggered susceptibility). The magnet, precisely. For debate d , each debater i carries a unit spin $s_i \in S^{n-1}$ ($n = 32$ whitened embedding dimensions; definition below). Sublattice A

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Pending.

Observables. Pending.

H21-debate-antiferromagnet · generated 2026-10-03 18:08 by infra/summaries/build_summaries.py · card hypotheses/H21-debate-antiferromagnet/README.md · ratings: not rated · DRAFT: auto stub, not yet written

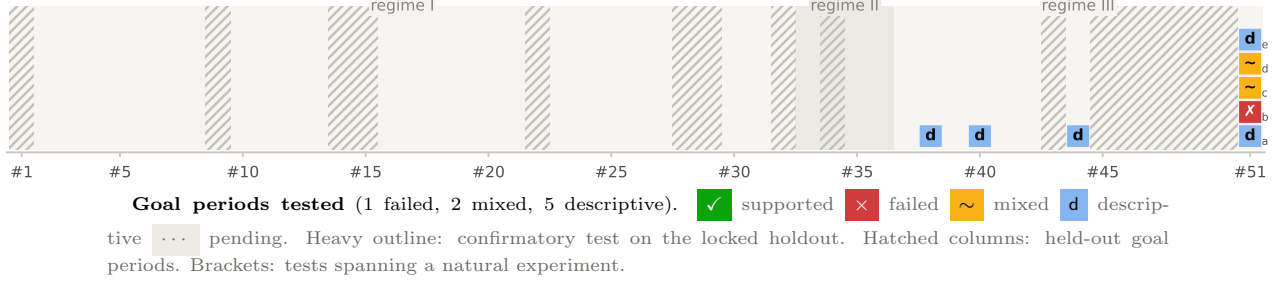
H22: Private, conflicting goals make #51 a spin glass

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 1/5 Usefulness 1.5/5

Bottom line. #51 is not a spin glass: private roles make it a random-field system with weak, mostly positive coupling, where rivals align rather than repel and the swarm never switches collectively.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

Do #51's private, partly conflicting roles make the swarm a spin glass? Tested on sub-periods 51a-e (45 non-holdout days) against shared-objective weeks #38, #40, #44, after synthetic validation (the balance index separates factions from random signs; SK and a heterogeneous ferromagnet cannot be told apart under common drive). Content couplings are real in 51b and 51c ($\rho_{\text{split}} = 0.46$, $p = 0.003$) but 85–94% positive ($\kappa = 4$ –10); the balance index is indeterminate. Same-role rivals co-move *more* than unrelated pairs ($T_{\text{SR}} = +0.048$, $p = 0.02$), the homophily rival. Positions persist ($M = 0.5$ – 0.75) without collective switching ($W \approx 1$). Contrast weeks are no less ferromagnetic; talk couplings sit at the noise floor. Holdout test (#51 tail) written, **not run**.

PHYSICAL MODEL

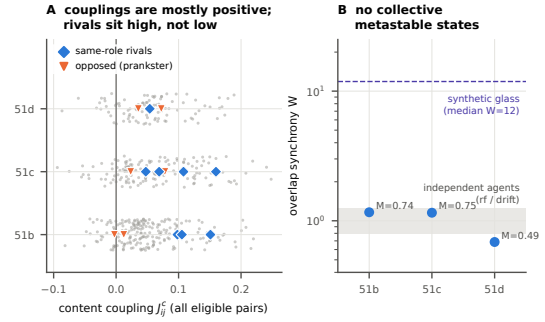
Models 01 (SK) and 11 ($O(n)$ spins). Spin $\mathbf{s}_i$: agent i 's whitened chat embedding per 30-min window ($n = 32$). Fields: day field $\mathbf{m}_d$; static field $\mathbf{H}_i$ (role, lab, style). J_{ij} : correlation of within-day fluctuations $\mathbf{x}_i, \mathbf{x}_j$ minus a cross-day surrogate; A, B, C : day folds. κ : SK ratio J_0/J ; τ_3 : gauge-invariant balance (SK ≈ 0 , factions ≈ 0.5); q : overlap of day-demeaned states $\hat{\delta}$; W : its synchrony beyond per-agent shifts.

$$P(\{\mathbf{s}\}) \propto \exp \left(\sum_{i < j} J_{ij} \mathbf{s}_i \cdot \mathbf{s}_j + \sum_i (\mathbf{m}_d + \mathbf{H}_i) \cdot \mathbf{s}_i \right)$$

$$J_{ij} = \text{corr}_w(\mathbf{x}_i, \mathbf{x}_j) - \text{corr}_w^{\text{surr}}(\mathbf{x}_i, \mathbf{x}_j)$$

$$\kappa = \sqrt{N} \bar{J} / \sigma_J, \quad \tau_3 = \text{tr}(J^A J^B J^C) / \text{tr}(J^A J^B)^{3/2}$$

$$q(d, d') = \frac{1}{N} \sum_i \hat{\delta}_{i,d} \cdot \hat{\delta}_{i,d'}$$



Observables. (a) Every eligible pair's content coupling in the counted #51 units (gray), with same-role rivals (blue) and Prankster vs Ethicist/Psychologist pairs (orange): mostly positive, rivals high. (b) Overlap synchrony W at the independent-agent level (gray band), far below a synthetic glass (dashed). High memory M without synchrony: agents hold their own positions; the swarm never jumps between collective states.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Practically, little. "Spin glass" was the exciting reading of mixed-motive swarms: frustrated, history-dependent, hard to steer. The data say something duller and more usable. Assigned roles act as fixed fields that pin each agent's topic, the shared room adds a weak positive pull, and same-objective agents drift together. For an operator, roles, not inter-agent conflict, set what such a swarm talks about, and there are no hidden collective states to escape. Caveat: embeddings measure topic, not stance, so strategic opposition could be invisible here. The diagnostics port to other logs; the spin-glass vocabulary is not earned.

H22-private-goals-spin-glass · generated 2026-10-03 18:08 by infra/summaries/build-summaries.py · card hypotheses/H22-private-goals-spin-glass/README.md · ratings: calibrated by coordinator 2026-10-04 (from H22 agent (suggested))

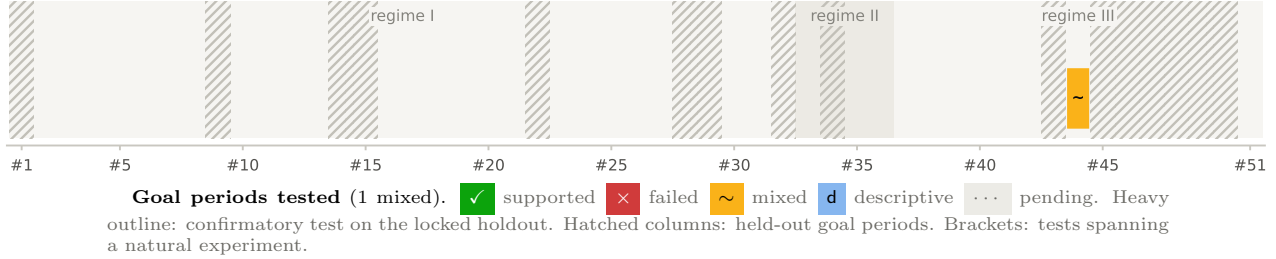
H23: Distilling the village into a fine-tuned leader copies its vocabulary but transforms its plans

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-03

Complete 40% Faithfulness 1.5/5 Usefulness 1.5/5

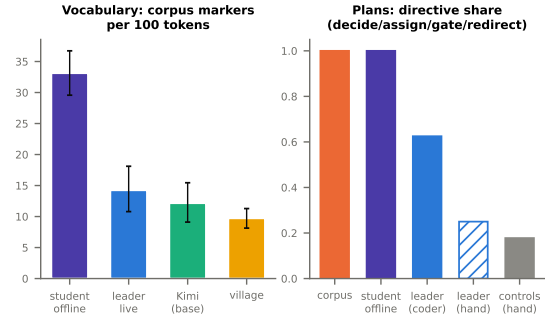
Bottom line. The fine-tuned leader carried a trace of its corpus’s addressing style into the room but none of its plans, which were not transmitted rather than transformed, and its corpus was mostly its own base model talking to itself.

level: hypothesis · pipeline checks 86% · updated 2026-10-03



WHAT HAPPENED

From the agents’ terminals: the deployed leader (agent 28 in #44, agent 30 in #45) is base Kimi K2.6 LoRA-tuned on 64 rows, 80% of them Kimi’s own replies to 20 synthetic village scenarios under a leadership prompt; no village chat was a target. On its 16 live #44 messages (G44, exploratory, not blind) vs 45 same-window messages: corpus-distinctive vocabulary 14.2 vs 10.0 per 100 tokens ($p = 0.011$), mostly agent addressing and markup; phrase copy n.s.; lexical copy information 0.012 bits per word. Plans: directive share 0.62 (hand-coded 0.25) vs the corpus’s 1.00; excess plan copy -0.04 vs a perfect-copy 0.23, transformation information at the null: *not transmitted*, not transformed. Not closer to the corpus than to its base in embeddings; base-echo and latest-chat rivals stand. #45 test written, **not run**.



Observables. (a) Corpus-distinctive n-grams per 100 tokens (95% CI): the offline student (sibling checkpoints on training-format prompts) sits far above the room; the live leader keeps about a fifth of that excess, near base Kimi in the same window ($n = 6$). (b) Directive share of plan acts: corpus and offline student 100%; the live leader (hatched: post hoc hand coding) sits near the room.

PHYSICAL MODEL

Model 08 (Kolchinsky & Corominas-Murtra), distillation variant: a channel from the corpus X to the leader’s live output Y on a shared alphabet (word n-grams, or 10 keyword-coded plan acts). Live message m is paired with the corpus rows whose scenario is nearest its context, $p_C(x | m)$. d : binary KL; r_x : probability of reproducing x ; a, b : rates at which the leader and untrained same-room speakers emit corpus n-grams; c : copy fraction.

$$I(X; Y) = I_{\text{copy}} + I_{\text{transform}}$$

$$I_{\text{copy}} = \sum_x p(x) d(r_x \| p_Y(x)) \mathbf{1}[r_x > p_Y(x)]$$

$$p(x, y) = \frac{1}{n} \sum_m p_C(x | m) \delta_{y, y_m}, \quad c = \sum_x p(x, x)$$

$$I_{\text{copy}}^{\text{lex}} = d(a \| b) \text{ bits per n-gram}$$

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For anyone distilling a swarm into a leader model, the lesson is mundane: SFT on a few dozen synthetic crisis scenarios carried the surface style (naming agents, markup, terse format) into deployment but not the decision policy, because live situations (routine project work) looked nothing like the training scenarios. That coverage failure is standard ML, not a swarm insight. The decomposition adds one useful distinction (not transmitted vs transformed) and a calibration (the perfect-copy reference), but 16 messages and a coder at $\kappa \approx 0.45$ measure little. The real surprise is the pipeline: the “village distillation” was the base model distilling itself.

H23-leader-distillation-copy · generated 2026-10-03 18:08 by infra/summaries/build_summaries.py · card hypotheses/H23-leader-distillation-copy/README.md · ratings: calibrated by coordinator 2026-10-04 (from H23 agent (suggested))

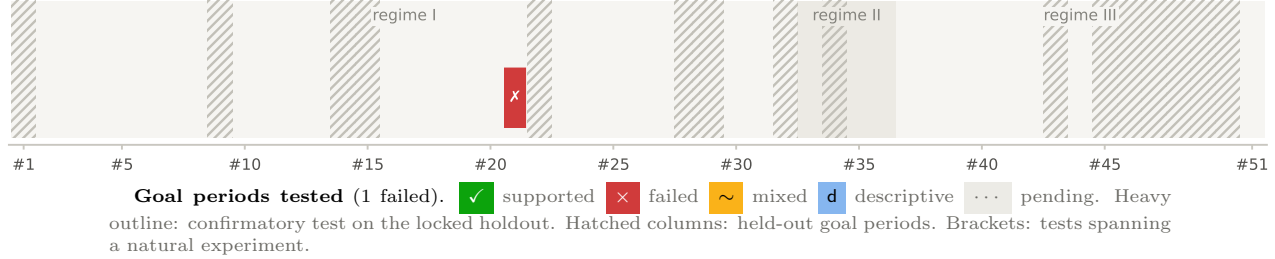
H24: In the forecast week (#21), switching from independent drafts to comparison switches the coupling on

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-03

Complete 40% Faithfulness 1/5 Usefulness 1.5/5

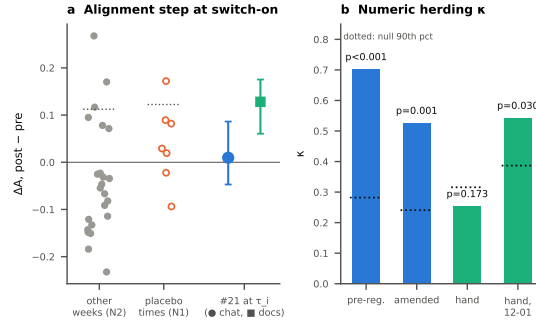
Bottom line. In the forecast week the coupling switch was real (teammate mentions jumped from 7% to 74%) but early, and it moved neither content alignment nor, after an audit, the numeric forecasts: the agents were aligned from the first hour.

level: hypothesis · pipeline checks 86% · updated 2026-10-03



WHAT HAPPENED

Goal period #21 (forecast AI; regime I; 8–9 agents) as a natural experiment. The switch-on was located from the record before any outcome. Agents’ session intentions and their own screen descriptions put each agent’s first reading of a teammate’s forecast 42–86 min into the week (median 50 min), not mid-week. The share of messages naming a teammate jumps there from 7% to 74%. The predicted alignment step did not appear: $\Delta A_{\text{res}} = +0.010 [-0.05, +0.09]$, inside both a within-week placebo and 23 other kickoff days. The synthetic gives ≈ 0.7 power at $\beta J_0/n \approx 0.55$. Content was already aligned before any reading ($A_{\text{res}} \approx 0.33$ vs. a rotation null of 0.03) and ramped over the week ($\rho = 0.73$). Numeric herding passed as pre-registered ($\kappa = 0.70$, $p < 0.001$) but was an extraction artifact: hand-verified values give $\kappa = 0.25$ ($p = 0.17$). Holdout (#14) not run.



Observables. (a) Change in residual alignment across the switch-on for #21 (blue: statements; green: documents; 90% CIs), against 23 other regime-I kickoff days at the same offset (gray) and placebo switch times inside #21 (orange). Dotted lines: 90th percentiles. (b) DeGroot pull κ under the pre-registered extractor, the amended extractor and hand-verified values; dotted: the independent-updating null, 90th percentile. The herding signal shrinks to noise once extraction errors are removed.

PHYSICAL MODEL

Model 11 (vector spins), with model 01’s Curie–Weiss analogue. Spin $\mathbf{s}_i$: agent i ’s whitened, unit-normalized statement embedding ($n = 32$). Field $h\hat{g}$: the goal and kickoff text. $\mathbf{h}_i$: agent style. Coupling J_0 : a mean-field pull toward the others, switched on at τ_i (the agent’s first reading of a teammate’s forecast). A : mean pairwise cosine, with $\hat{g}$ removed for A_{res} . Numeric forecasts x_i (AGI by 2035, SI by 2050, p(doom)) are soft spins with a DeGroot pull κ .

$$\mathbf{s}_i(t) \propto h\hat{g} + \mathbf{h}_i + \xi_i(t) + J_0 \Theta(t - \tau_i) \mathbf{m}_{-i}(t)$$

$$A = \frac{2}{N(N-1)} \sum_{i < j} \hat{\mathbf{v}}_i \cdot \hat{\mathbf{v}}_j, \quad \beta J_0/n = 1 - \frac{1}{1 + (N-1)A_{\text{res}}}$$

$$\Delta x_i = \kappa (\bar{x}_{-i} - x_i) + \epsilon_i, \quad \kappa = 0 \text{ for independent updating}$$

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Mostly a cautionary tale. Telling LLM agents to “form independent views, then compare” did not buy independence: their content was aligned before anyone read anyone, from the shared prompt, instructions and prior. Comparison then produced talk about divergence, and some literal copying of numbers, not a measurable alignment jump. For anyone aggregating swarm forecasts, an independent-first-draft phase is no diversity guarantee: most of the diversity worth protecting is gone at kickoff. The reusable lessons are methodological. Regex-extracted numbers manufacture herding through regression to the mean, and snapshot coupling estimates absorb unmodelled fields. One week, $N = 7$, no ground truth: a pilot.

H24-forecast-coupling-switch · generated 2026-10-03 18:08 by infra/summaries/build-summaries.py · card hypotheses/H24-forecast-coupling-switch/README.md · ratings: calibrated by coordinator 2026-10-04 (from H24 agent (suggested))